



Dolby Voice Huddle Version 5.0

Administrator's Guide

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Notices

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Contents

1 Introduction to this guide.....	7
1.1 About this documentation.....	8
1.2 Related documentation.....	8
1.3 Problem reports.....	8
1.4 Documentation feedback.....	9
2 Dolby Voice Huddle Overview.....	10
2.1 Dolby Voice Huddle Features and Benefits.....	11
2.2 Dolby Voice Huddle hardware components.....	11
2.2.1 Dolby Voice Huddle Camera.....	11
2.3 Dolby Voice Huddle software.....	11
2.5 Operation modes of Dolby Voice Huddle	12
2.6 Supported IP PBX call control platforms.....	12
3 Preparation and requirements.....	14
3.1 Conference Room requirements.....	15
3.1.1 Room size and acoustics.....	15
3.1.2 Power.....	15
3.1.3 Display.....	15
3.1.4 Cabling.....	16
3.1.5 Lighting conditions and image quality.....	16
3.2 Network requirements.....	16
3.2.1 DHCP and DNS servers.....	16
3.2.2 Firewall.....	16
3.2.3 VLAN.....	16
3.2.4 Network ports used.....	17
3.2.5 802.1X.....	17
3.2.6 HTTP proxy.....	17
3.2.7 NTP server.....	17
3.2.8 Bandwidth planning.....	18
3.2.9 QoS limitation.....	18
4 Deploying Dolby Voice Huddle	19
4.1 Deployment overview.....	20
4.2 Provisioning Overview.....	20
4.3 Deploying the plug and play provisioning method.....	21
4.3.1 Testing a plug and play provisioning deployment.....	21
4.4 Deploying local setup.....	21
4.4.1 Deploying the automatic provisioning method.....	22
4.4.2 Deploying the manual provisioning method.....	23
4.5 Handling network issues during provisioning.....	23
4.5.1 Configuring 802.1X during out-of-the-box setup.....	24
4.5.2 Configuring 802.1X after out-of-the-box setup.....	25

4.5.3 Primary and secondary VLANs.....	26
4.5.4 Changing the VLAN setting.....	29
4.5.5 Manually specifying a provisioning server.....	30
4.6 Setting up individual Dolby Voice Huddle systems.....	30
5 Configuring Dolby Voice Huddle features.....	32
5.1 Configuration parameters.....	33
5.2 Setting and viewing configuration parameters.....	33
5.3 Configuring certificates.....	34
5.3.1 Device certificate.....	34
5.3.2 Adding certificate authority certificates.....	35
5.3.3 Viewing certificate authority certificates status.....	35
5.3.4 Trusted certificates and issuers.....	36
5.4 Changing display settings.....	39
5.5 Changing audio input and output devices.....	39
5.6 Configuring the Dolby Voice Camera.....	39
5.6.1 Adjusting the camera for lighting conditions.....	40
5.6.2 Vertically flipping the camera position.....	40
5.6.3 Resetting camera settings.....	40
5.6.4 Creating and editing presets using PTZ control.....	41
5.7 Configuring the Dolby Voice Huddle Controller.....	41
5.8 Configuring IP PBX.....	41
5.8.1 Prerequisites for configuring IP PBX.....	41
5.8.2 Support for SIP high availability.....	42
5.8.3 Configuring IP PBX using the web interface.....	43
5.8.4 Configuring IP PBX using the Dolby Conferencing Console.....	44
5.8.5 Adjusting codec preferences.....	46
5.8.6 Enabling Opus on Dolby Voice Huddle.....	47
5.8.7 Configuring voice activation detection.....	48
5.8.8 Configuring IP PBX for SIP URI dialing.....	48
5.8.9 SIP video configuration.....	49
5.9 Configuring preferences.....	50
5.9.1 Disabling the recent-calls list.....	50
5.9.2 Enabling user feedback.....	51
5.9.3 Configuring call forwarding.....	51
5.9.4 Enabling automatic answer.....	52
5.9.5 Enabling do not disturb.....	53
5.9.6 Setting the language.....	54
5.9.7 Setting the time and date display.....	54
5.10 Configuring contact directories.....	57
5.10.1 Corporate directory.....	57
5.10.2 Local directory.....	57
5.10.3 Configuring the corporate directory.....	58
5.10.4 Creating a local contact directory from the provisioning server.....	59
5.10.5 Resetting the directory.....	59
5.10.6 Resetting call logs.....	59
5.11 Configuring the time server.....	60

5.12 Calibrating audio.....	60
5.13 Disabling the reboot prompt.....	61
6 Administering Dolby Voice Huddle.....	62
6.1 Changing the administrative password.....	63
6.2 Accessing the Dolby Voice Huddle web interface.....	63
6.3 Administering with the diagnostic shell.....	64
6.3.1 Diagnostic shell.....	64
6.3.2 Opening the diagnostic shell.....	65
6.4 Viewing the status menu.....	66
6.5 Setting the Dolby Voice Huddle operational mode.....	66
6.6 Viewing and managing event logs.....	66
6.6.1 Event logging.....	66
6.6.2 Log levels.....	67
6.6.3 Audio logging.....	67
6.6.4 syslog logging.....	68
6.6.5 Enabling HDMI logging.....	69
6.6.6 Viewing log files.....	69
6.7 Initiating Reprovisioning.....	70
6.8 Enabling power saving mode.....	71
6.9 Changing network settings.....	71
6.10 Enabling Privacy block.....	72
6.11 Call and audio statistics.....	73
6.12 Enabling WebRTC tracing.....	73
6.13 Rebooting Dolby Voice Huddle.....	74
6.14 Resetting to factory defaults.....	74
6.15 Updating the Dolby Voice Huddle software.....	75
6.16 Configuring HDMI input audio.....	76
6.16.1 Disabling HDMI input audio.....	76
6.17 Finding the system serial number and software version.....	76
6.18 Finding the serial number and software version.....	76
7 Troubleshooting.....	77
7.1 Report to Dolby.....	78
7.2 Service QR code.....	78
7.3 Web administration portal access over Wi-Fi.....	78
7.4 Enabling screenshots.....	79
7.5 Dolby Voice Camera issues and solutions.....	80
7.6 Dolby Voice Huddle system issues and solutions.....	80
7.6.1 Power and start-up issues.....	80
7.6.2 Dolby Voice Huddle power saving issues and solutions.....	80
7.6.3 Network access issues.....	81
7.6.4 Authentication issues.....	81
7.6.5 HDMI input issues.....	81
7.6.6 Software update issues.....	82
7.6.7 Video quality issues.....	83
7.6.8 Corporate directory issues.....	84

7.6.9 Fatal error issues.....	84
7.7 Display issues and solutions.....	85
7.8 HDMI input issues.....	85
7.9 CEC issues.....	86
8 Error Messages.....	87
8.1 Dolby Voice Console error messages.....	88
8.2 Provisioning service error messages.....	91
8.3 Dolby Voice Huddle application error messages.....	93
8.4 Dolby Voice Huddle touch controller application error messages.....	95
Glossary.....	99

Introduction to this guide

Dolby Voice Huddle is an integrated software and hardware conferencing solution for huddle spaces and small conference rooms.

- [About this documentation](#)
- [Related documentation](#)
- [Problem reports](#)
- [Documentation feedback](#)

1.1 About this documentation

This documentation is for IT personnel who intend to deploy and administer one or more Dolby Voice Huddle systems.

Users of this documentation should be familiar with:

- Basics of computer networking and Linux administration
- IP private branch exchange (PBX) call control platform used by your organization
- Conferencing service provider functionality used by your organization

This documentation provides information about:

- Dolby Voice Huddle architecture
- Preparation and requirements for setting up your Dolby Voice Huddle systems
- Provisioning methods and procedures
- Configuring the system using the Dolby Voice Huddle user interface (UI) and web interface
- Handling errors

1.2 Related documentation

Review all of the Dolby Voice Huddle documents to ensure that the system is set up correctly and that you have the best conferencing experience possible.

These documents are available:

- *Dolby Voice Huddle Quick Start Guide*

This guide describes the contents of the Dolby Voice Huddle package, how to assemble the system, and how to connect the system to the network. The quick start guide is included in the package. It is also available from the Dolby Voice Huddle support pages.

- *Dolby Conferencing Console Administrator's Guide*

This guide describes using Dolby Conferencing Console as a provisioning server for Dolby Voice Huddle.

- *Dolby Voice Product Compatibility Guide*

This guide describes the compatibility relationships between the various Dolby Voice products.

1.3 Problem reports

When you report problems to your conferencing service provider, be prepared to provide basic system information, log files, and a full description of the issue.

- Basic system information:
 - Serial number
 - Software version
- Any relevant log files showing the problem occurring, or anything that might be relevant in the log. Did anything unusual occur? Did the system generate any high-severity log messages? If so, attach an extract. If you have not enabled logging, enable it for the system experiencing the issue, reproduce the issue, and provide the logs.
- Description of the issue:
 - When did the problem occur? How often does it occur? Is there any pattern or trend to the occurrence?
 - What was the scope of the problem? How many users did it affect? Was there any pattern or trend to the affected users?
 - Any symptoms observed.

- Any error messages displayed on the controller user interface or web interface.
- Any obvious triggers. (What was the user doing before the error message displayed?)
- If the issue was reproducible. (Does it happen regularly or after certain tasks are performed?) If you can reproduce the problem, detail how.
- How widespread the issue appears to be. (Does it affect only a single user or the whole organization?)
- Any other information you have that might assist in understanding the issue.
- What operating system and version are being used by the client? What browser and version are being used by the client?
- What kind of Internet connection is the client using (ADSL/cable/wireless/mobile)?
- What other observations have you made? Is there anything else you think might assist us in identifying the root cause of the problem?

1.4 Documentation feedback

If you have comments or feedback about this documentation, send us an email at dolbyvoicedocs@dolby.com.

2

Dolby Voice Huddle Overview

Dolby Voice Huddle is an integrated hardware and software conferencing system for huddle spaces and small conference rooms.

- [Dolby Voice Huddle Features and Benefits](#)
- [Dolby Voice Huddle hardware components](#)
- [Dolby Voice Huddle software](#)
- [Operation modes of Dolby Voice Huddle](#)
- [Supported IP PBX call control platforms](#)

2.1 Dolby Voice Huddle Features and Benefits

Dolby Voice Huddle offers these capabilities:

- Advanced video and audio conferencing when integrated with conferencing service providers that support Dolby Voice technologies.
- Traditional public switched telephone network (PSTN) calling features when paired with IP PBX call control platforms.

Benefits include:

- 4K Self-Monitoring, Analysis, and Reporting Technology (SMART) camera with 110° wide-angle lense captures all meeting participants.
- Dynamic leveling boosts quiet voices while distinguishing speech from noise to make all voices clear at a consistent volume.
- 3D Audio to capture and map voices to create the feeling of an in-person meeting
- Intelligent Acoustics™ regularly calibrates and optimizes audio for your space, whether using your TV speakers or a sound bar.
- High dynamic range (HDR) video mapping continuously balances exposure levels for improved visibility no matter the lighting conditions.
- HDMI input to easily share screens and content.
- Simplified setup and automated deployment.
- Easy conference joining.
- Remote device management for monitoring and troubleshooting.

2.2 Dolby Voice Huddle hardware components

The Dolby Voice Huddle system includes these hardware components:

- Dolby Voice Camera
- Dolby Voice Huddle Controller
- Dolby Voice Huddle Expansion Mic

To complete the conferencing system, you connect one high-definition television or monitor (display).

2.2.1 Dolby Voice Huddle Camera

Dolby Voice Huddle Camera is a SMART camera with 4K resolution that is embedded in the Dolby Voice Huddle unit.

It includes these features:

- 110° wide-angle field of view
- Digital pan-tilt-zoom (PTZ) capability
- HDR capability

The camera includes a mounting system that permits mounting on a display (TV or monitor). See the *Dolby Voice Huddle Quick Start Guide* for more information.

2.3 Dolby Voice Huddle software

The Dolby Voice Huddle system software includes the Dolby Voice Camera firmware, and the Dolby Voice Huddle Controller firmware.

Dolby Voice Huddle manages software updates automatically for the camera and controller.

The software handles system functions, manages multiple user interfaces, and provides a platform for running conferencing service provider applications.

System functions include:

- Management of the protocol stack, the DSP, the UI, and the network interaction
- Voice over Internet Protocol (VoIP) signaling for a range of voice telephony functions using Session Initiation Protocol (SIP) signaling for call setup and control
- Advanced audio signal processing for communications using a group of audio codecs
- Industry-standard security techniques for ensuring that all provisioning, signaling, and media transactions are authenticated and encrypted
- Flexible provisioning methods to support various deployment scenarios

User interfaces for administration include:

- Touch-screen interface on the Dolby Voice Huddle
- Web interface
- Command-line interface with the diagnostic shell

2.5 Operation modes of Dolby Voice Huddle

The Dolby Voice Huddle system can integrate with the conferencing service provider as a conference endpoint, register to an IP PBX call control platform as a SIP audio-only endpoint, or do both simultaneously.

The operation mode determines which features are supported. For example, if the operation mode is set to **Conferencing Only Mode**, it cannot be configured to connect to an IP PBX, and features such as dialing a local extension, call forwarding, and call transferring are unavailable.

If the operation mode is set to **Dual Mode**, the system can connect to both an IP PBX and a conferencing service enabled for Dolby Voice.

For more information, see the configuration guide for your call control platform.

2.6 Supported IP PBX call control platforms

An IP PBX provides audio communications through a TCP/IP protocol stack for endpoints on internal networks and interconnects with a PSTN for telephony connections for external endpoints.

Dolby Voice Huddle supports these IP PBX call control platforms:

- Avaya Aura Platform
- BroadSoft BroadWorks
- Cisco Unified Communications Manager
- Unify OpenScape Enterprise Express

Standard SIP call control platform

Dolby Voice Huddle is a SIP-compliant endpoint and interoperates with call control platforms designed to comply with RFC 3261. For Transport Layer Security (TLS) transport, the system is limited by the existing security configuration. Keep in mind:

- If the system transport is configured as AUTO, TLS transport is attempted only if the system fails to connect to Transmission Control Protocol (TCP) and User Datagram Protocol (UDP) transports.
- The CA certificate of the SIP server must be downloaded to the system through provisioning.
- TLS 1.0, 1.1, and 1.2 are supported. TLS 1.2 is the preferred protocol, but version 1.0 or 1.1 can be used, depending on peer support.
- Because the system acts as a client, the server determines whether or not mutual authentication is required. (Normally, SIP servers are configured only with client authentication.)
- Certificate revocation lists (CRLs) are not supported.

- Simple Certificate Enrollment Protocol (SCEP) and Online Certificate Status Protocol (OCSP) are not supported.
- The system device certificate cannot be replaced at this time.
- If you try to set up IP PBX with explicit TLS transport instead of using the **Auto** transport feature, you must set the transport port explicitly. If you change your configuration from TLS to **Auto**, you must set the port back to the correct default port.

Contact your network administrator to find out which port is used for the TLS connection in your environment.

When TLS is operational on the system, Secure Real-time Transport Protocol (SRTP) is enabled; when TLS is not operational on the system, SRTP is disabled. By default, if TLS is enabled, SRTP is set to Mandatory. Dolby Voice Huddle offers a predefined list of cipher suites.

3

Preparation and requirements

To successfully deploy Dolby Voice Huddle, you must first complete some tasks to satisfy both room requirements and network requirements.

Before you begin, read the *Dolby Voice Huddle Quick Start Guide*.

- [Conference Room requirements](#)
- [Network requirements](#)

3.1 Conference Room requirements

There are specific requirements for the conference room, including size, acoustics, lighting conditions, and necessary equipment.

3.1.1 Room size and acoustics

Before you install Dolby Voice Huddle, make sure that the room does not exceed the maximum recommended room size.

From an acoustics point of view, choosing a Dolby Voice Huddle solution based on the size of the room is important because it ensures natural and effective conversational flow for the usual seating positions in the room. Participant behavior and environmental factors influence the overall voice quality in the room and impact your experience; these factors include:

- Participant seating position
- Participant voice volume
- Ambient noise level
- Reverberation level
- Floor, wall, and ceiling materials
- Ceiling fans
- Projector placement

When Dolby Voice Huddle is deployed in acoustically challenging rooms, we recommend testing the setup to determine whether the actual results meet your requirements.

3.1.2 Power

The Dolby Voice Huddle system requires a nearby power outlet, 110–240 V, 50–60 Hz.

 **Note:** Be aware that the system powers on as soon as it is plugged in, without the need to press the power button. Therefore, avoid connecting the system to power until you have connected all hardware components.

3.1.3 Display

Dolby Voice Huddle supports single-display configurations, where the display is either a TV or monitor.

This table summarizes display requirements.

Table 1: Display Requirements

Rear clearance	If you mount the Dolby Voice Camera on the display, allow 3 inches (7.6 cm) clearance from the front of the display bezel.
Positioning for camera field of view	If you intend to mount the Dolby Voice Camera on the display, keep in mind that the camera has a 110-field of view. Consider also the position of the display relative to any whiteboard.

Table 1: Display Requirements (continued)

Resolution	Supported resolutions: • 1080p • 4K (ultra-high definition, 3840 × 2160, 16:9)
Sound Quality	For maximum sound quality, it is best to use displays with low latency and minimum sound processing features (less sound processing features mean less delay). High quality speakers and speaker placement are important for sound quality. For example, if the unit is mounted on top of the display, the down-firing speakers will project the audio away from the satellite microphone. Also, do not place the unit, display, and satellite microphone together on a small table.

3.1.4 Cabling

An unshielded twisted-pair CAT6 cable is required to connect the Dolby Voice Huddle.

When the Dolby Voice Huddle is mounted on the display, we recommend that you enclose the CAT6 and high-definition video cables in a cable-management system.

3.1.5 Lighting conditions and image quality

We recommend that you optimize lighting conditions for best image quality.

For example, ensure that the room is well lit. If it is not practical to adjust lighting conditions, you can adjust these image settings with the controller UI:

- Brightness
- Color intensity
- Contrast
- HDR (High Dynamic Range) Video Mapping

3.2 Network requirements

Network connectivity must meet certain minimum requirements, and must be available in all locations where you intend to deploy Dolby Voice Huddle.

3.2.1 DHCP and DNS servers

Both Dynamic Host Configuration Protocol (DHCP) and Domain Name System (DNS) servers must be available to support Dolby Voice Huddle.

If a DHCP server is not available, you can set a static IP address. However, DHCP addressing is preferred, because it provides a simpler setup experience.

3.2.2 Firewall

Dolby Voice Huddle requires TCP port 443 for the domain *dolbyvoice.com*. Check with your conferencing service provider to determine other firewall requirements, including WebRTC port requirements.

3.2.3 VLAN

The Dolby Voice Huddle system requires internet access to connect to the conferencing service. In some environments, voice and data traffic are handled by different virtual LAN (VLAN)s.

3.2.4 Network ports used

The network port used by the Dolby Voice Huddle depends on protocol, data type, and whether it is sent by UDP or TCP.

Table 2: Network ports used

Port number	Protocol	Data	UDP or TCP
22	SSH	Diagnostics	TCP
80 ^[a]	Hypertext Transfer Protocol (HTTP)	Not applicable	TCP
443	Hypertext Transfer Protocol Secure (HTTPS)	Not applicable	TCP
5060	SIP	SIP signaling	UDP or TCP
5061	SIP over TLS	Secure signaling	TCP
16400–16431 ^[b]	RTP/RTCP	Media, audio	UDP

[a] Port 80 is always redirected to port 443.

[b] Real-time Transport Protocol (RTP) ports are always even numbers; RTP Control Protocol (RTCP) ports are always odd numbers. The numbers always start from 16,400/16,401 and progress to 16,402/16,403 and so forth until they reach the end of the range. At this point, they revert to 16,400/16,401 and start over again.

3.2.5 802.1X

If your network uses 802.1X authentication, then you must understand the restrictions placed on devices that do not have 802.1X credentials. You must obtain 802.1X credentials for every Dolby Voice Huddle system.

You can customize these Dolby Voice Huddle supported authentication methods:

- EAP-PEAP: Requires the device to validate the CA certificate, and the user to enter a user name and/or password, for network authentication.
- EAP-TLS: Provides for certificate-based and mutual authentication of the client and the network. It relies on client-side and server-side certificates to perform authentication.
- EAP-MD5: Requires the creation of an MD5 hash of a random value from the network, and the user to enter a password, to authenticate.
- EAP-TTLS: The only supported secondary authentication method is MSCHAPv2.
- EAP-FAST with automatic proxy automatic configuration (PAC) provisioning: The system supports anonymous or authenticated in-band PAC provisioning. Manual PAC provisioning and the "do not use PAC" provisioning modes are not supported.

3.2.6 HTTP proxy

Dolby Voice Huddle can use an HTTP proxy server to connect to services on the Internet.

Dolby Voice Huddle supports Web Proxy Automatic Discovery (WPAD) and PAC.

Static proxy server configuration is not supported.

3.2.7 NTP server

Dolby Voice Huddle can use a network time server, so your DHCP server should supply Network Time Protocol (NTP) server information.

If your DHCP server does not supply this information, you can configure NTP server information manually. If the desired NTP server is not on your network, ensure that your firewall rules allow connection to *time.nist.gov* and *pool.ntp.org*.

3.2.8 Bandwidth planning

Dolby Voice Huddle supports the H.264 video codec and the Opus 8 kHz (20ms) audio codec. Check with your conferencing service provider for audio and video bandwidth requirements.

Normally a minimum of 4 Mbps, for both upload and download, is recommended.

3.2.9 QoS limitation

Dolby Voice Huddle does not support paramater configurations for Quality of Service (QoS).

Deploying Dolby Voice Huddle

As the Dolby Voice Huddle system administrator, you work with IT to deploy the system in your organization.

- [Deployment overview](#)
- [Provisioning Overview](#)
- [Deploying the plug and play provisioning method](#)
- [Deploying local setup](#)
- [Handling network issues during provisioning](#)
- [Setting up individual Dolby Voice Huddle systems](#)

4.1 Deployment overview

Deploying Dolby Voice Huddle in your organization consists of these tasks.

1. Select a method by which to provision your Dolby Voice Huddle systems.
2. Complete some deployment tasks to support the selected method.
3. Install and provision a Dolby Voice Huddle system to test the deployment. Handle network and other issues as needed.
4. After a successful test, set up Dolby Voice Huddle in your conference rooms and huddle spaces.

4.2 Provisioning Overview

As a Dolby Voice Huddle administrator, you must understand the Dolby Voice Huddle provisioning process. Provisioning refers to performing out-of-the-box (OOB) setup of the Dolby Voice Huddle system.

During provisioning, the system downloads configuration files, updates software if needed, and downloads the conferencing service provider application with which users interact. Provisioning can also be initiated on demand and configured to recur at regular intervals.

There are multiple ways to provision Dolby Voice Huddle:

- Plug and play provisioning method is a fast and easy way to set up Dolby Voice Huddle using your conferencing service provider's provisioning server. It makes trials and small-scale deployments of Dolby Voice Huddle easier for administrators. This is the preferred method.
- Local setup method is for users who want to use an on-site provisioning server, such as Dolby Conferencing Console. There are two local setup methods:
 - Automatic provisioning method
 - Manual provisioning method

When the Dolby Voice Huddle system starts up for the first time out of the box, it follows this provisioning procedure:

1. Contact the DHCP server to obtain an IP address and discover network resources, including automatic discovery of a proxy server.
2. Query the DNS server for the host name of a possible on-premises Dolby Conferencing Console.
3. Contact Dolby Voice Console in the cloud, and if Dolby Conferencing Console information was found, pass along that information.

Dolby Voice Console then determines which provisioning server the system should use: the on-premises Dolby Conferencing Console, if any (automatic provisioning method), or the conferencing service provider's provisioning server (plug and play provisioning method).

If the system is registered with Dolby Voice Console for conferencing service provider provisioning, and an on-premises Dolby Conferencing Console is present, the latter takes precedence. If the system is not registered with Dolby Voice Console, the system displays a screen asking the user to select a conferencing service provider.

Typically, your conferencing service provider or distributor registers your systems with Dolby Voice Console.

4. Download the provisioning information from the selected provisioning server, guide the user through the setup experience defined by the conferencing service provider, and reboot.

Errors might prevent the system from discovering a provisioning server. The most common errors are network issues, including:

- Physical connectivity issues
- HTTP/HTTPS proxy server issues
- 802.1X authentication issues

- VLAN issues
- NTP issues

The system displays manual setup screens to enable you to correct errors, for example, by entering network parameters or missing credentials. After you correct the errors, the system then attempts to complete the provisioning process. At certain points in this process, you can choose to skip the remainder of the steps and start the manual provisioning method.

For more information on Dolby Conferencing Console, see *Dolby Conferencing Console Administrator's Guide*.

4.3 Deploying the plug and play provisioning method

The plug and play provisioning method requires coordination with your conferencing service provider.

Prerequisites

Ensure that you have completed all preparations and requirements, keeping in mind that all Dolby Voice Huddle systems must connect to a provisioning server outside the firewall.

Procedure

1. Confirm with your conferencing service provider that you have adequate service accounts to deploy Dolby Voice Huddle.
2. If your network uses 802.1X authentication, obtain credentials for every Dolby Voice Huddle system.

4.3.1 Testing a plug and play provisioning deployment

After you deploy the plug and play provisioning method, test it by provisioning a single Dolby Voice Huddle system.

About this task

See the *Dolby Voice Huddle Quick Start Guide* for more information.

Procedure

1. Unbox the Dolby Voice Huddle system.
2. Mount the Dolby Voice Camera to the display and connect them using an HDMI cable.
3. Connect the Dolby Voice Camera to the Dolby Voice Huddle Controller.
4. Connect the Dolby Voice Camera to the Dolby Voice Huddle Expansion Mic.
5. Connect the Dolby Voice Camera to the network, using an ethernet cable.
6. Connect the power cable to the Dolby Voice Huddle Camera.
7. Power up the Dolby Voice Huddle system.

4.4 Deploying local setup

You can configure Dolby Voice Huddle for local setup if you require a configuration that is more advanced than the plug and play provisioning method.

You can deploy Dolby Voice Huddle using local setup for these scenarios:

- You have more than six Dolby Voice Huddles and want to control the Dolby Voice Huddle configuration, such as IP PBX configuration.
- You want an aggregated report for both Dolby Voice Huddle and IP PBX usage in your rooms.

There are two local setup methods:

- Automatic provisioning method that simplifies deployment and minimizes configuration errors for large or multiple-site deployment scenarios. Uses an on-premises Dolby Conferencing Console, which you manage, to provision each Dolby Voice Huddle system.
- Manual provisioning method used if Dolby Voice Huddle did not successfully discover a provisioning server, or if you want manual control.

4.4.1 Deploying the automatic provisioning method

We recommend using the automatic provisioning method. Using this method, the out-of-the-box setup wizard can discover the Dolby Conferencing Console or the provisioning server on your network. The automatic provisioning method is an on-premises, bulk method that simplifies deployment and minimizes configuration errors for multiple-site deployment scenarios.

Prerequisites

- Verify that you have administrative access to the DNS server.
- Ensure that your Dolby Conferencing Console is reachable on the network.
- Ensure that you have completed all preparations and requirements.

Procedure

1. Prepare the configuration files on Dolby Conferencing Console. Create profiles and device pools. See the *Dolby Conferencing Console Administrator's Guide* for details.
2. Create a Domain Name System service record (DNS SRV) for your on-premises Dolby Conferencing Console.
3. If your network uses 802.1X authentication, obtain credentials for every Dolby Voice Huddle system.

Testing an automatic provisioning deployment

After you deploy the automatic provisioning method, test it by provisioning a single Dolby Voice Huddle system.

About this task

See the *Dolby Voice Huddle Quick Start Guide* for more information.

Procedure

1. Unbox the Dolby Voice Huddle system.
2. Mount the Dolby Voice Camera to the display and connect them using an HDMI cable.
3. Connect the Dolby Voice Camera to the Dolby Voice Huddle Controller.
4. Connect the Dolby Voice Camera to the Dolby Voice Huddle Expansion Mic.
5. Connect the Dolby Voice Camera to the network, using an ethernet cable.
6. Connect the power cable to the Dolby Voice Huddle Camera.
7. Power up the Dolby Voice Huddle system.

DNS SRV record syntax for the automatic provisioning method

When this DNS SRV record is present, Dolby Voice Huddle can discover the on-premises Dolby Conferencing Console for provisioning.

Syntax

The DNS SRV record is an entry in the forward zone file and has this syntax:

```
_service._protocolName IN SRV priority weight portNO target subdns.domain.com
```

`_service._protocolName`

`_dccs._tcp` if you want the system to connect to Dolby Conferencing Console with HTTPS, or `_dcc._tcp` if you want the system to connect to Dolby Conferencing Console with HTTP. If records for both of these protocols are present, HTTPS takes precedence.

`priority`

Numeric. Lower gets first priority.

`weight`

Numeric. Higher gets used more often.

`portNO`

The port used.

443: The typical port used by the HTTPS provisioning server.

80: The typical port used by the HTTP provisioning server.

`subdns.domain.com`

The fully qualified domain name (FQDN) of the Dolby Conferencing Console on your network.

4.4.2 Deploying the manual provisioning method

Use the manual provisioning method if the system did not discover the Dolby Conferencing Console or a provisioning server, or if you want manual control. The manual provisioning method begins when you skip the selection of a service provider during system setup.

About this task

The first part of out-of-the-box setup involves network setup. You can use the setup wizard to connect the system to the network manually, if desired. After you complete this portion and have connected to the network, the next step is connecting to the provisioning server.

Procedure

1. Connect the Dolby Voice Huddle system to an ethernet port on the network using an ethernet cable.
The setup wizard displays on the controller UI.
2. If a provisioning server is not found, tap **Manual setup**.
3. From the **Select language** screen, choose a language. If needed, scroll up or down to find the language.
4. From the **Select time zone** screen, choose a time zone. If needed, scroll up or down to find the time zone.
5. **Optional:** From the **Network setup** screen, you can change your virtual LAN settings and proxy settings, if needed.
6. If challenged for 802.1X authentication, tap **Confirm certificate**, provide the valid credentials, and then follow the instructions in [Deploying with 802.1X authentication](#).
You will be presented with a certificate, depending on the protocol method you selected.
7. Wait for the controller to connect to the network.

What to do next

After you have connected to the network, continue through the setup wizard to connect to the provisioning server.

4.5 Handling network issues during provisioning

Network issues might prevent provisioning from completing. Here are the common network setup tasks that you might need to perform during provisioning.

About this task

If the provisioning process halts at the **Language** screen, this indicates that Dolby Voice Huddle was unable to connect to a provisioning server. From the **Network Configuration** screen, you can correct network issues and then complete the provisioning process.

4.5.1 Configuring 802.1X during out-of-the-box setup

If your network requires 802.1X during the network setup portion of the setup wizard, you are required to complete 802.1X setup. By default, 802.1X authentication is disabled during out-of-the-box setup.

About this task

You can use configuration parameters, the controller UI, or the web interface to enable 802.1X authentication.

Once you have enabled 802.1X authentication for Dolby Voice Huddle, you can customize the protocol methods you want to use.

Procedure

- Set the configuration parameter as described in the following table.

Configuration parameter name	Configuration parameter description
Network.EapOl.Enabled	Enables/disables 802.1X authentication. Set to false.

- To enable 802.1X authentication from the web interface:

 **Note:** If you need a Wi-Fi connection to access your device, see *Web administration portal access over Wi-Fi* in the *Troubleshooting* chapter.

- Access the Dolby Voice Huddle web interface.
 - Navigate to **Settings > Network > IEEE 802.1X**.
 - Toggle **Enable IEEE 802.1X Authentication** to turn 802.1X authentication on.
- To enable 802.1X authentication from the controller UI:
 - Access the Dolby Voice Huddle settings menu.
This may be different for each service provider.
 - Enter the admin password.
 - Tap **Network configuration > 802.1X**
 - Tap **Administrative settings > Network configuration > 802.1X**
 - Toggle **Enable 802.1X Authentication** to turn 802.1X authentication on.
After enabling 802.1X, you can customize protocol methods, by selecting **Methods** from the **802.1X authentication** screen.
 - Enable the protocol methods you want to use, and tap the confirm button.
The **Network setup** screen returns, showing successful authentication.
 - Enter the authentication credentials.
 - Confirm the certificate.
 - Tap the confirm button to continue provisioning.

What to do next

After you complete these steps, return to the provisioning task that you were working on before you were prompted to confirm this certificate.

Related information

[Configuring 802.1X after out-of-the-box setup](#) on page 25

[Primary and secondary VLANs](#) on page 26

[Changing the VLAN setting](#) on page 29

[Manually specifying a provisioning server](#) on page 30

4.5.2 Configuring 802.1X after out-of-the-box setup

You can configure 802.1X authentication after you have completed the out-of-the-box setup.

About this task

If a guest VLAN is not configured, you must use the controller UI to enable 802.1X authentication.

If a guest VLAN is permitted, you can use the controller UI and web UI to enable 802.1X authentication.

Procedure

- Set the configuration parameters as described in the following table.

Configuration parameter description	Configuration parameter name
Enables or disables 802.1X authentication	Network.Eapol.Enabled
Sets the Institute of Electrical and Electronics Engineers (IEEE) 802.1X user identity	Network.Eapol.Identity
Sets the IEEE 802.1X password	Network.Eapol.Password
Sets the 802.1X authentication protocol method	Network.Eapol.EnableEapFast
	Network.Eapol.EnableEapMd5
	Network.Eapol.EnableEapPeap
	Network.Eapol.EnableEapTls
	Network.Eapol.EnableEapTtls

- Use the web interface:
 - Access the Dolby Voice Huddle web interface.
 - Navigate to **Settings > Network > IEEE 802.1X**.
 - Toggle **Enable IEEE 802.1X Authentication** to turn 802.1X authentication on.

Go to **Settings > Network > IEEE 802.1X**, and enter the parameters or toggle the buttons to enable the protocol methods.
- To enable 802.1X authentication from the controller UI:
 - Access the Dolby Voice Huddle settings menu.

This may be different for each service provider.
 - Enter the admin password.
 - Tap **Network configuration > 802.1X**
 - Tap **Administrative settings > Network configuration > 802.1X**
 - Toggle **Enable 802.1X Authentication** to turn 802.1X authentication on.

After enabling 802.1X, you can customize protocol methods, by selecting **Methods** from the **802.1X authentication** screen.
 - Enable the protocol methods you want to use, and tap the confirm button.

The **Network setup** screen returns, showing successful authentication.
 - Enter the authentication credentials.
 - Confirm the certificate.
 - Tap the confirm button to continue provisioning.

Related information

[Configuring 802.1X during out-of-the-box setup](#) on page 24

[Primary and secondary VLANs](#) on page 26

[Changing the VLAN setting](#) on page 29

[Manually specifying a provisioning server](#) on page 30

4.5.3 Primary and secondary VLANs

The Dolby Voice Huddle system requires internet access to connect to the conferencing service. In some environments, voice and data traffic are handled by different VLANs.

If the voice VLAN lacks Internet connectivity, modify firewall policies to allow the system to connect to your conferencing service. Alternatively, you can set up the system on the data VLAN. If neither of these solutions works for you, then you need a dual-VLAN setup. After you enable dual VLANs, communication over the Internet with servers for your conferencing service provider passes through the secondary VLAN. With separate VLANs, the system has two IP addresses, one for each VLAN, but the same media access control (MAC) address.

When the secondary VLAN is enabled, the primary VLAN is designated as the voice VLAN, and the secondary VLAN is designated as the data VLAN. When the secondary VLAN is disabled, the system can be placed on either the data VLAN or the voice VLAN. If your organization has a strict policy on segregating voice and data VLANs, and you want to configure the system for both IP PBX and conferencing service, enable the secondary VLAN.

Check the following to verify VLAN status:

DNS Server Status

Displays status of DNS for the voice and data VLAN.

The **Status** screens with voice and data VLAN settings are shown in these figures.

Figure 1: Status screen displaying Voice VLAN settings

Network

Host Name

VCS8000-E0001178

Ethernet

Ethernet Duplex

full

Ethernet Speed (Mbps)

1000

Total Packets Received

13122

Total Packets Transmitted

6648

DHCP

DHCP Server Address

10.0.0.1

Virtual LAN

VLAN ID In Use

0

Alarms and Status

DHCP Status

✓

DNS Server Status

✓

Default Gateway Status

✓

Hostname Resolvable

!

IEEE 802.1X Authentication Method

Unknown

IEEE 802.1X Authentication Phase 2 Method

Unknown

IEEE 802.1X Authentication Status

●

IEEE 802.1X Client Certificate

Dolby Factory Installed Certificate

IP Address Conflict

Link Layer State

✓

NTP Server Discovery

✓

PAC Status

●

PDS Connection Status

●

Voice VLAN Status

●

WPAD Status

●

Figure 2: Status screen displaying Data VLAN settings

+
Network

-
Network - Secondary VLAN

Alternative DNS Server Address
DNS Server Address
Default Gateway
Domain Name
IP Address
Subnet Mask

+
DHCP

-
Status

DHCP Status

DNS Server Status

Default Gateway Status

IP Address Conflict

PAC Status

Secondary VLAN Status

WPAD Status

Network traffic is allocated onto different VLANs based on the traffic type, as listed in this table.

Table 3: Traffic-type-to-network-interface mapping

Traffic type	Network interface	Comment
DHCP	Primary and secondary	Works independently on both
WPAD	Primary and secondary	If dual VLAN is enabled, the secondary is supported.
PAC	Primary and secondary	If dual VLAN is enabled, the secondary is supported.
DNS ^[a]	Primary and secondary	DNS queries made on the network interface where the DNS requestor application runs
Conferencing services	Secondary	
Lightweight Directory Access Protocol (LDAP)	Secondary	
Web server	Primary and secondary	
SSH server	Primary and secondary	
Provisioning	Secondary	
SIP call	Primary	

[a] If the primary VLAN shares the same DNS settings, then DNS traffic runs only on the primary virtual LAN.

Dual VLAN limitations

Dual VLAN has these limitations:

- In attached mode, if dual VLAN is enabled, a voice VLAN must be assigned to Dolby Voice Huddle .
- The secondary VLAN must be untagged on the Dolby Voice Hub.
- On Dolby Voice Huddle, provisioning requests can only be performed on the secondary VLAN.
- If dual VLAN is enabled for Dolby Voice Huddle in a 802.1x environment and the Cisco ethernet switch, you must use multi-host mode on the Cisco switch.
- If dual VLAN is enabled, the primary VLAN proxy is disabled.
- Secondary VLAN IP assignment is only supported through DHCP, user cannot assign static IP to the secondary VLAN.

Voice VLAN checklist

Check the following to determine whether your network is ready for dual VLAN deployment.

Data VLAN checklist

If both checklist tests work

If both tests work, you can proceed with the dual VLAN configuration.

 **Note:** When the dual-VLAN feature is used in conjunction with a Cisco switch port with 802.1X authentication enabled, the Ethernet port on the switch must be configured to Multiple-Hosts mode. If the port is set to Multidomain Authentication or Multiauthentication mode, dual VLAN does not work on the system.

Related information

[Configuring 802.1X during out-of-the-box setup](#) on page 24

[Configuring 802.1X after out-of-the-box setup](#) on page 25

[Changing the VLAN setting](#) on page 29

[Manually specifying a provisioning server](#) on page 30

4.5.4 Changing the VLAN setting

Change the VLAN setting if your VLAN 0 (the default) has no internet access and thus prevents provisioning from proceeding. Or if provisioning succeeds, you can change the VLAN later if you choose not to use the default VLAN.

About this task

This procedure assumes that a VLAN issue occurred, preventing provisioning from completing, and that you have viewed the error message on the **Setup unsuccessful** screen and tapped **Next** to go to the **Network setup** screen.

Procedure

1. Access the Dolby Voice Huddle settings menu.
This may be different for each service provider.
2. Enter the admin password.
3. Tap **Network Configuration > Ethernet > Manual VLAN ID**.
4. Enter the **Manual VLAN ID**.
5. Tap **Enter**.
6. Tap the confirm button.

Related information

[Configuring 802.1X during out-of-the-box setup](#) on page 24

[Configuring 802.1X after out-of-the-box setup](#) on page 25

[Primary and secondary VLANs](#) on page 26

[Manually specifying a provisioning server](#) on page 30

4.5.5 Manually specifying a provisioning server

During out-of-the-box setup, you can specify a provisioning server to use by manually entering the server address.

About this task

There are a few ways to manually specify a provisioning server:

- With the controller UI
- With the web interface

Using the controller UI

1. Navigate to the settings menu.

This may be different for each service provider.

2. Tap Administrative Settings.
3. Enter the password.
4. Tap **Provisioning Server**.
5. Enable **Manual Server Configuration**.
6. Tap **Provisioning Server Type**
7. Select the server type.
8. Tap the confirm button to save the changes.

Using the web interface

1. Access the Dolby Voice Huddle web interface.
2. Go to **Settings > Provisioning > Server**.
3. Enable **Manual Server Configuration**.
4. Specify the **Provisioning Server Type**.
5. Enter the **Provisioning Server Address**.
6. Click **Save**.

If the server configuration is valid, provisioning proceeds with the specified server.

Related information

[Configuring 802.1X during out-of-the-box setup](#) on page 24

[Configuring 802.1X after out-of-the-box setup](#) on page 25

[Primary and secondary VLANs](#) on page 26

[Changing the VLAN setting](#) on page 29

4.6 Setting up individual Dolby Voice Huddle systems

After deploying and testing the desired provisioning method, you can install and set up individual Dolby Voice Huddle systems.

About this task

See the *Dolby Voice Huddle Quick Start Guide* for more information.

Procedure

- 1.** Unbox the Dolby Voice Huddle system.
- 2.** Mount the Dolby Voice Camera to the display and connect them using an HDMI cable.
- 3.** Connect the Dolby Voice Camera to the Dolby Voice Huddle Controller.
- 4.** Connect the Dolby Voice Camera to the Dolby Voice Huddle Expansion Mic.
- 5.** Connect the Dolby Voice Camera to the network, using an ethernet cable.
- 6.** Connect the power cable to the Dolby Voice Camera.
- 7.** Power up the Dolby Voice Huddle system.

5

Configuring Dolby Voice Huddle features

As the Dolby Voice Huddle administrator, you can configure many features of the system, including network settings, corporate directory features, display and camera settings, calling features, and more. You configure these features by setting and changing configuration parameters.

- [Configuration parameters](#)
- [Setting and viewing configuration parameters](#)
- [Configuring certificates](#)
- [Changing display settings](#)
- [Changing audio input and output devices](#)
- [Configuring the Dolby Voice Camera](#)
- [Configuring the Dolby Voice Huddle Controller](#)
- [Configuring IP PBX](#)
- [Configuring preferences](#)
- [Configuring contact directories](#)
- [Configuring the time server](#)
- [Calibrating audio](#)
- [Disabling the reboot prompt](#)

5.1 Configuration parameters

A large set of configuration parameters controls Dolby Voice Huddle features and settings. The system receives and stores the majority of configuration parameters from the provisioning server. You can manually set other configuration parameters using one of the interfaces that the system provides.

A detailed description of all configuration parameters, along with the default or permitted values for those parameters, is available in the **Help** section of the web interface.

 **Note:** If a configuration parameter is set by the provisioning server, you cannot change it manually.

Default values

If a configuration parameter has not been set manually and is not present in the configuration files on the provisioning server, a default value is used.

Configuration parameters requiring reboot

Some configuration parameters require a reboot when changed. If the system requires a reboot, a reboot prompt displays on the controller user interface and web interface. If you do not cancel the reboot within 20 seconds, the system reboots automatically. Active calls are not interrupted; the reboot is delayed until those calls have ended. You can turn off the reboot prompt if desired.

Device pools and profiles

With Dolby Conferencing Console, you can easily assign multiple configuration settings to multiple devices in a single operation. You can define device pools, which are collections of Dolby Voice Huddle systems, and you can create profiles, which are collections of configuration parameters and their values. You can then apply profiles to pools. See the *Dolby Conferencing Console Administrator's Guide* for details.

 **Note:** If you use the plug and play provisioning method, you cannot configure features for a pool of systems, because your systems are provisioned by your conferencing service provider. You can still configure features one system at a time (for example, you can configure IPPBX settings, which are not provisioned by your conferencing service provider's provisioning service).

5.2 Setting and viewing configuration parameters

There are a few ways to set and view configuration parameters.

Procedure

-  **Note:** If a setting is dimmed, it means that this configuration parameter is set by the provisioning server and cannot be changed manually.

- Use the web interface.

 **Note:** If a setting is dimmed, it means that this configuration parameter is set by the provisioning server and cannot be changed manually.

- Use the **configure** command in the diagnostic shell to set a parameter.
For example, to configure the system to automatically answer calls, enter this command:

```
configure Preferences.AutoAnswerEnabled true
```

To set the logging level to DEBUG for the SIP protocol, enter this command:

```
configure Logging.Level.Sip DEBUG
```

- Use the **show configuration** command in the diagnostic shell to view a parameter.

For example, to see how long a web interface or diagnostic shell session remains active before login is required again, enter this command:

```
show configuration Authentication.Local.SessionTimeout
```

To show all configuration parameters, enter this command:

```
show configuration all
```

To get a list of related configuration parameters, such as all parameters related to PBX, enter the first portion or first two portions of the parameter name, up to the period, and press Tab.

```
show configuration Sip.Pbx. (Tab)
```

```
Sip.Pbx.CallFailoverMode
Sip.Pbx.CallFailoverTimeout
Sip.Pbx.Codecs
Sip.Pbx.ContentShareEnabled
Sip.Pbx.Domain
Sip.Pbx.EnableP2PSIPVideo
Sip.Pbx.EnableURIDialingProxy
Sip.Pbx.InBandDTMF
Sip.Pbx.OnlyAcceptCallServerEvents
Sip.Pbx.OutboundProxy1
Sip.Pbx.OutboundProxy1Port
Sip.Pbx.OutboundProxy2
Sip.Pbx.OutboundProxy2Port
Sip.Pbx.RegistrationTimeout
Sip.Pbx.SipUriScheme
Sip.Pbx.TLSCertNameMatchingPolicy
Sip.Pbx.URIDialingEnabled
Sip.Pbx.URIDialingProxy
Sip.Pbx.URIDialingTransport
```

5.3 Configuring certificates

The collection of certificate authority (CA) and device certificates installed on the Dolby Voice Huddle system is a security feature and is used for authentication with features such as HTTPS provisioning, LDAP TLS connection, TLS SIP signaling encryption, 802.1X, and conferencing services that support Dolby Voice.

The system can trust certificates issued by widely recognized certificate authorities when trying to establish a connection to a server. To establish new trust relationships, you can add new certificates to the system through Dolby Conferencing Console or the web interface.

5.3.1 Device certificate

A device certificate is installed on each Dolby Voice Huddle system at the factory.

The device certificate is an X.509 digital certificate signed by the Dolby root certificate authority. The digital certificate and associated private key are stored on the system in its nonvolatile memory as part of the manufacturing process. For more information on digital certificates, see www.ietf.org/html.charters/pkix-charter.html and IETF RFC 2459.

Issuer name	Expiration date	CA Certificate download link
.Dolby.Comms.Devices.CA.1	Expires before 2050	https://certificates.dolbyvoice.com/dolby_comms_devices_ca1_chain.cer

Related information

[Adding certificate authority certificates](#) on page 35

[Viewing certificate authority certificates status](#) on page 35

[Trusted certificates and issuers](#) on page 36

5.3.2 Adding certificate authority certificates

A certificate that is issued by a certificate authority establishes a point of trust between two entities, for example, the Dolby Voice Huddle system and the LDAP directory or the provisioning server.

If you want the system to trust any CAs, you must add their certificates to the system in a CA certificate directory. This file must be in Privacy-Enhanced Electronic Mail (PEM) encoded format.

You can add CA certificates from:

- Dolby Conferencing Console

For details, see the *Dolby Conferencing Console Administrator's Guide*.

- The web interface

1. Access the Dolby Voice Huddle web interface.
2. From the **Tools > Certificates > Import CA certificates**.

A standard file-selection dialog box displays.

3. Select the CA certificate file and click **Open**.



Note: Certificates imported through the web interface are deleted after a factory reset. You must re-import the certificates.

Related information

[Device certificate](#) on page 34

[Viewing certificate authority certificates status](#) on page 35

[Trusted certificates and issuers](#) on page 36

5.3.3 Viewing certificate authority certificates status

With the CA certificates status screen, you can view the number of certificates that are expired, are about to expire, or are not yet valid.

About this task

All certificates are validated every time the certificate store changes, as well as every 24 hours if scheduled provisioning is enabled. Warnings and error messages are available from the controller UI and the web interface. If a certificate expires, the **Expired** count increases on the CA certificates status screen. If a certificate is set to expire within seven days, the **Expiring soon** count increases on the CA certificates status screen. You must replace the certificate before expiry, or certain features will stop working on the system.

Procedure

- To view CA certificates status from the controller interface:
 - a) Using the Dolby Voice Huddle Controller, navigate to the settings menu.
 - b) Navigate to **About > CA Certificates**.
- To view CA certificates status from the web interface:
 - a) From the **Status** tab, click **Dolby Voice Huddle**, then **Device**, then **CA Certificates**.
 - b) **Optional:** To see a list of certificates, including a list of expired certificates, click the link next to **CA Certificates Details**.



Tip: A list of expired certificates appears before the full list of certificates. You can select the hexadecimal number in the error message, copy it to the clipboard, and then use your browser's find command to find the matching certificate in the full list of certificates.

Related information

[Device certificate](#) on page 34

[Adding certificate authority certificates](#) on page 35

[Trusted certificates and issuers](#) on page 36

5.3.4 Trusted certificates and issuers

Dolby Voice Huddle trusts the certificates listed here.

Table 4: Trusted certificates

Certificate name	Issuer
*.lobby.pp.iconf.net	Entrust Certification Authority - L1K
Amazon Root CA 1	Starfield Services Root Certificate Authority - G2
Amazon Root CA 1	Self-signed
Amazon Root CA 2	Self-signed
Amazon Root CA 3	Self-signed
Amazon Root CA 4	Self-signed
Baltimore CyberTrust Root	Baltimore CyberTrust Root
Baltimore CyberTrust Root	Baltimore CyberTrust Root
C=US/O=Starfield Technologies, Inc./OU=Starfield Class 2 Certification Authority	http://www.valicert.com// emailAddress=info@valicert.com
C=US/O=The Go Daddy Group, Inc./OU=Go Daddy Class 2 Certification Authority	C=US/O=The Go Daddy Group, Inc./OU=Go Daddy Class 2 Certification Authority
C=US/O=VeriSign, Inc./OU=Class 2 Public Primary Certification Authority - G2/OU=(c) 1998 VeriSign, Inc. - For authorized use only/OU=VeriSign Trust Network	C=US/O=VeriSign, Inc./OU=Class 2 Public Primary Certification Authority - G2/OU=(c) 1998 VeriSign, Inc. - For authorized use only/OU=VeriSign Trust Network
C=US/O=VeriSign, Inc./OU=Class 3 Public Primary Certification Authority - G2/OU=(c) 1998 VeriSign, Inc. - For authorized use only/OU=VeriSign Trust Network	C=US/O=VeriSign, Inc./OU=Class 3 Public Primary Certification Authority - G2/OU=(c) 1998 VeriSign, Inc. - For authorized use only/OU=VeriSign Trust Network
Cybertrust Public SureServer SV CA	Baltimore CyberTrust Root
DigiCert Assured ID Root CA	DigiCert Assured ID Root CA
DigiCert Assured ID Root G2	DigiCert Assured ID Root G2
DigiCert Assured ID Root G3	DigiCert Assured ID Root G3
DigiCert Global Root CA	DigiCert Global Root CA
DigiCert Global Root G2	DigiCert Global Root G2
DigiCert Global Root G3	DigiCert Global Root G3
DigiCert High Assurance EV Root CA	DigiCert High Assurance EV Root CA
DigiCert SHA2 High Assurance Server CA	DigiCert High Assurance EV Root CA
DigiCert Trusted Root G4	DigiCert Trusted Root G4
DOLBY-PSM	DOLBY-PSM
Entrust Root Certification Authority	Entrust Root Certification Authority
Entrust Root Certification Authority - G2	Entrust Root Certification Authority - G2
Entrust.net Certification Authority (2048)	Entrust.net Certification Authority (2048)

Table 4: Trusted certificates (continued)

Certificate name	Issuer
GeoTrust Global CA	GeoTrust Global CA
GeoTrust Global CA 2	GeoTrust Global CA 2
GeoTrust Primary Certification Authority	GeoTrust Primary Certification Authority
GeoTrust Primary Certification Authority - G3	GeoTrust Primary Certification Authority - G3
GeoTrust Universal CA	GeoTrust Universal CA
GeoTrust Universal CA 2	GeoTrust Universal CA 2
GlobalSign	GlobalSign
GlobalSign Root CA	GlobalSign Root CA
Go Daddy Root Certificate Authority - G2	Go Daddy Root Certificate Authority - G2
Google Internet Authority G3	GlobalSign
Microsoft Root Certificate Authority 2011	Microsoft Root Certificate Authority 2011
Microsoft Secure Server CA 2011	Microsoft Root Certificate Authority 2011
O=RSA Security Inc/OU=RSA Security 2048 V3	O=RSA Security Inc/OU=RSA Security 2048 V3
Starfield Root Certificate Authority - G2	Starfield Root Certificate Authority - G2
Starfield Services Root Certificate Authority - G2	C=US/O=Starfield Technologies, Inc./OU=Starfield Class 2 Certification Authority
Symantec Class 3 Secure Server CA - G4	VeriSign Class 3 Public Primary Certification Authority - G5
TC TrustCenter Class 2 CA II	TC TrustCenter Class 2 CA II
TC TrustCenter Universal CA I	TC TrustCenter Universal CA I
thawte Primary Root CA	thawte Primary Root CA
thawte Primary Root CA - G3	thawte Primary Root CA - G3
Thawte Universal CA Root	Thawte Universal CA Root
VeriSign Class 3 Public Primary Certification Authority - G5	VeriSign Class 3 Public Primary Certification Authority - G5
VeriSign Universal Root Certification Authority	VeriSign Universal Root Certification Authority

Related information

[Device certificate](#) on page 34

[Adding certificate authority certificates](#) on page 35

[Viewing certificate authority certificates status](#) on page 35

TLS cipher suites

The Dolby Voice Huddle offers a predefined list of cipher suites. This list cannot be changed at this time.

 **Note:** The current SRTP implementation supports the AES_CM_128_HMAC_SHA1_80 and AES_CM_128_HMAC_SHA1_32 cipher suites. It does not support the F8_128_HMAC_SHA1_8 cipher suite.

Table 5: TLS cipher suites

Cipher	Cipher suites
0x2f	TLS_RSA_WITH_AES_128_CBC_SHA
0x30	TLS_DH_DSS_WITH_AES_128_CBC_SHA
0x31	TLS_DH_RSA_WITH_AES_128_CBC_SHA
0x35	TLS_RSA_WITH_AES_256_CBC_SHA
0x36	TLS_DH_DSS_WITH_AES_256_CBC_SHA
0x37	TLS_DH_RSA_WITH_AES_256_CBC_SHA
0x3c	TLS_RSA_WITH_AES_128_CBC_SHA256
0x3d	TLS_RSA_WITH_AES_256_CBC_SHA256
0x3e	TLS_DH_DSS_WITH_AES_128_CBC_SHA256
0x3f	TLS_DH_RSA_WITH_AES_128_CBC_SHA256
0x41	TLS_RSA_WITH_CAMELLIA_128_CBC_SHA
0x42	TLS_DH_DSS_WITH_CAMELLIA_128_CBC_SHA
0x43	TLS_DH_RSA_WITH_CAMELLIA_128_CBC_SHA
0x84	TLS_RSA_WITH_CAMELLIA_256_CBC_SHA
0x85	TLS_DH_DSS_WITH_CAMELLIA_256_CBC_SHA
0x86	TLS_DH_RSA_WITH_CAMELLIA_256_CBC_SHA
0x9c	TLS_RSA_WITH_AES_128_GCM_SHA256
0x9d	TLS_RSA_WITH_AES_256_GCM_SHA384
0xa0	TLS_DH_RSA_WITH_AES_128_GCM_SHA256
0xa1	TLS_DH_RSA_WITH_AES_256_GCM_SHA384
0xa4	TLS_DH_DSS_WITH_AES_128_GCM_SHA256
0xa5	TLS_DH_DSS_WITH_AES_256_GCM_SHA384
0xc004	TLS_ECDH_ECDSA_WITH_AES_128_CBC_SHA
0xc005	TLS_ECDH_ECDSA_WITH_AES_256_CBC_SHA
0xc009	TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA
0xc00a	TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA
0xc00e	TLS_ECDH_RSA_WITH_AES_128_CBC_SHA
0xc00f	TLS_ECDH_RSA_WITH_AES_256_CBC_SHA
0xc013	TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA
0xc014	TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA
0xc023	TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA256
0xc024	TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA384
0xc025	TLS_ECDH_ECDSA_WITH_AES_128_CBC_SHA256
0xc026	TLS_ECDH_ECDSA_WITH_AES_256_CBC_SHA384
0xc027	TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA256
0xc028	TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA384

Table 5: TLS cipher suites (continued)

Cipher	Cipher suites
0xc029	TLS_ECDH_RSA_WITH_AES_128_CBC_SHA256
0xc02a	TLS_ECDH_RSA_WITH_AES_256_CBC_SHA384
0xc02b	TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256
0xc02c	TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384
0xc02d	TLS_ECDH_ECDSA_WITH_AES_128_GCM_SHA256
0xc02e	TLS_ECDH_ECDSA_WITH_AES_256_GCM_SHA384
0xc02f	TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256
0xc030	TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384
0xc031	TLS_ECDH_RSA_WITH_AES_128_GCM_SHA256
0xc032	TLS_ECDH_RSA_WITH_AES_256_GCM_SHA384

5.4 Changing display settings

Using the web interface, you can set the preferred display resolution and, in a dual-display configuration, swap images between the displays.

Procedure

1. Access the Dolby Voice Huddle web interface.
2. Navigate to **Settings > Displays**
3. Change **Preferred resolution** to one recommended by your conferencing service provider.
Dolby Voice Huddle uses that resolution if supported by the displays. The system falls back to a lower resolution if the displays do not support the selected resolution.

5.5 Changing audio input and output devices

Using the web interface, you can manually change audio input and output devices. By default, the **Audio Device Selection** option is set to **Automatic** and audio input and output device selection is performed automatically at initialization. Audio input and output device manual selection options are dependent on paired or available devices.

Procedure

1. Access the Dolby Voice Huddle web interface.
2. Under the **Settings** tab, expand **Audio** and select **Manual** for **Audio Device Selection**.
3. Manually select your audio device by modifying these settings:

Option	Description
Input Audio Device	Select Native Microphone , Satellite Microphone , or Dolby Conference Phone .
Output Audio Device	Select HDMI Output or Dolby Conference Phone .

5.6 Configuring the Dolby Voice Camera

Configure Dolby Voice Camera to take advantage of some of the advanced features of Dolby Voice Huddle.

Dolby Voice Camera operates in Room Mode, which captures a full view of the room, using the entire 110-degree field of view.

The camera can be configured with electronic PTZ through the Dolby Voice Huddle Controller.

5.6.1 Adjusting the camera for lighting conditions

Use the controller UI to adjust the quality of the image captured by the camera.

About this task

You can adjust the brightness, contrast, and color intensity (saturation) of the image. You can also turn on the HDR feature, which you can use to compensate for bright window light in the camera field of view. These adjustments are set to their default values when you clear user preferences.

 **Tip:** For a room with even lighting and without an outside-facing window, we recommend turning HDR off.

Procedure

1. Using the Dolby Voice Huddle Controller, navigate to **Settings > Dolby Voice Camera > Camera View** screen.
2. Adjust the sliders and optionally turn HDR on or off.

Related information

[Vertically flipping the camera position](#) on page 40

[Resetting camera settings](#) on page 40

[Creating and editing presets using PTZ control](#) on page 41

5.6.2 Vertically flipping the camera position

You can vertically flip the camera position using the web interface or through the command-line interface.

Procedure

- From the web interface:
 - a) Using the web interface, select **Settings > Dolby Voice Camera > Flip image vertically**.
 - b) Select **On**.
 - c) Click **Save**.
- From the command-line interface, enter:
`DolbyCli>configure DVCamera.VerticalFlip <string>(false)`

The parameter values are true and false.

Related information

[Adjusting the camera for lighting conditions](#) on page 40

[Resetting camera settings](#) on page 40

[Creating and editing presets using PTZ control](#) on page 41

5.6.3 Resetting camera settings

Use the Dolby Voice Huddle Controller to reset the camera settings and revert it to factory default.

Procedure

1. Using the Dolby Voice Huddle Controller, navigate to **Settings > Dolby Voice Camera > Reset Camera Settings**.
The **Reset Camera Settings** screen appears.

2. Click on the confirm button to reset to factory defaults.

Related information

[Adjusting the camera for lighting conditions](#) on page 40
[Vertically flipping the camera position](#) on page 40
[Creating and editing presets using PTZ control](#) on page 41

5.6.4 Creating and editing presets using PTZ control

You can create and edit up to three presets and assign specific pan, tilt, and zoom settings for each one.

Procedure

1. Using the Dolby Voice Huddle Controller, navigate to **Settings > Dolby Voice Camera > Camera View**.
2. Select a preset and click the edit button.
3. Click on **View Name** to rename the preset.
4. Click **Enter**.
5. Drag the rectangle to adjust the pan and tilt positions.
6. Use the -/+ buttons or the sliding scale to adjust the zoom.
7. Click **Done** to confirm your changes.

Related information

[Adjusting the camera for lighting conditions](#) on page 40
[Vertically flipping the camera position](#) on page 40
[Resetting camera settings](#) on page 40

5.7 Configuring the Dolby Voice Huddle Controller

Use Dolby Voice Huddle Controller to configure these features of Dolby Voice Huddle:

- Preferences
- Telephony
- Dolby Voice Camera
- General Settings
- Administrative Settings

5.8 Configuring IP PBX

To use Dolby Voice Huddle to make IP audio calls, you must configure IP PBX settings.

About this task

Use the web interface or the Dolby Conferencing Console to configure the account settings for your IP PBX and register the device. This involves modifying the **Account**, **Server**, and **Credential** settings in the **IP PBX** section.

Results

After you finish setting up the device for IP PBX, the phone icon on the home screen no longer appears dimmed.

5.8.1 Prerequisites for configuring IP PBX

Configuring IP PBX for the Dolby Voice Huddle system requires that you have information about the SIP domain name and the outbound proxy server address.

Your IP PBX service provider or IT department may provide the following information:

- One of the following SIP server/service connection information:
 - SIP domain name only.
 - SIP outbound proxy server address and/or SIP domain name. There may be two addresses for redundancy.
 - SIP domain name and SIP transport type.
 - SIP domain name, outbound proxy server address, and SIP transport type.
- SIP user account (required).
- SIP authentication user name; also known as the digest authentication user name (required).
- SIP authentication user password (required).

Related information

[Support for SIP high availability](#) on page 42

[Configuring IP PBX using the web interface](#) on page 43

[Configuring IP PBX using the Dolby Conferencing Console](#) on page 44

[Adjusting codec preferences](#) on page 46

[Enabling Opus on Dolby Voice Huddle](#) on page 47

[Configuring voice activation detection](#) on page 48

[Configuring IP PBX for SIP URI dialing](#) on page 48

[SIP video configuration](#) on page 49

5.8.2 Support for SIP high availability

You can configure IP PBX settings for the Dolby Voice Huddle to provide uninterrupted server availability and DNS-based redundancy.

The Dolby Voice Huddle supports the following for SIP high availability:

- Server failover/failback
- DNS redundancy
- DNS-based load balancing
- NAPTR record support

Server failover has two modes: `ACTIVE_SERVER` and `PREFERRED_ORDER`. The `ACTIVE_SERVER` mode switches/fails over to a secondary server after ending the call. The `PREFERRED_ORDER` mode attempts to fail over to the preferred list of IP addresses during the call. You can specify the server failover mode in the `Sip.Pbx.CallFailoverMode` field.

If your IP PBX service provider configures the Dolby Voice Huddle with only the SIP domain, the Dolby Voice Huddle performs the DNS SRV query on the domain host name. After four attempts to register to the IP address, the Dolby Voice Huddle attempts to register to the next IP in the DNS response.

If your IP PBX service provider configures the Dolby Voice Huddle with the SIP domain name and outbound proxy server address configured, the Dolby Voice Huddle performs the DNS SRV query on the outbound proxy server. After four attempts to register to the IP address, the Dolby Voice Huddle attempts to register to the next IP in the DNS response.

You can specify NAPTR record as the transport type in the `Sip.Account.TransportType` field.

The following table lists the various high-availability configuration options available:

Table 6: High availability deployment configuration options

IP PBX server configuration options	Outcome
Sip.Pbx.Domain as IP address	SIP requests are sent to the IP address configured in the Sip.Pbx.Domain field.
Sip.Pbx.Domain as <i>hostname</i> :5060	SIP requests are sent to the IP address resolving the host name configured in the Sip.Pbx.Domain field.
Sip.Pbx.Domain as <i>hostname</i> and Sip.Pbx.OutboundProxy1 as IP address	SIP requests are sent to the IP address configured in the Sip.Pbx.OutboundProxy1 field.
Sip.Pbx.Domain as <i>hostname</i> , Sip.Pbx.OutboundProxy1 as IP address, and Sip.Pbx.OutboundProxy2 as IP address	SIP requests are sent to the IP address configured in the Sip.Pbx.OutboundProxy1 field or the IP address configured in the Sip.Pbx.OutboundProxy2 field.
Sip.Pbx.Domain as <i>hostname</i> or <i>hostname</i> :0 only and Sip.Account.TransportType set to NAPTR record	NAPTR record query (to discover transport) followed by SRV query (to discover port) followed by A query.
Sip.Pbx.Domain as <i>hostname</i> or <i>hostname</i> :0 only and Sip.Account.TransportType set to AUTOSRV	SRV query (to discover port), followed by A query.
Sip.Pbx.Domain as <i>hostname</i> , Sip.Pbx.OutboundProxy1 as <i>hostname</i> , Sip.Pbx.OutboundProxy1Port as 0, and Sip.Account.TransportType as AUTO	SRV query (to discover port) followed by A query.
Sip.Pbx.Domain as <i>hostname</i> , Sip.Pbx.OutboundProxy1 as <i>hostname</i> , Sip.Pbx.OutboundProxy1Port as 0, Sip.Pbx.OutboundProxy2 as <i>hostname</i> , Sip.Pbx.OutboundProxy2Port as 0, and Sip.Account.TransportType as AUTO.	SRV query (to discover port) followed by A query.

Related information

- [Prerequisites for configuring IP PBX](#) on page 41
- [Configuring IP PBX using the web interface](#) on page 43
- [Configuring IP PBX using the Dolby Conferencing Console](#) on page 44
- [Adjusting codec preferences](#) on page 46
- [Enabling Opus on Dolby Voice Huddle](#) on page 47
- [Configuring voice activation detection](#) on page 48
- [Configuring IP PBX for SIP URI dialing](#) on page 48
- [SIP video configuration](#) on page 49

5.8.3 Configuring IP PBX using the web interface

You can use the web interface to configure IP PBX for the Dolby Voice Huddle system.

Procedure

1. Access the Dolby Voice Huddle web interface.
2. On the **Settings** tab, navigate to **IP PBX Settings > Account** and enter this information:
 - In the **Display Name** field, enter the device name.
 - In the **Extension number/Address** field, enter the extension or SIP user account.
 - If the dial-in number is different from the extension number/address, enter the dial-in number in the **Display Number** field.
 - If your IP PBX service provider has specified the transport type, choose the appropriate setting from **Transport Type**.
3. On the **Settings** tab, navigate to **IP PBX Settings > Server** and enter this information:

- If a SIP domain is available, enter it in the **SIP Domain Name** field.
- If an outbound proxy server is available, enter it in the **Primary Call Server/Outbound Proxy** field. To enable redundancy, enter an alternative outbound proxy server address in the **Secondary Call Server/Outbound Proxy** field.
- If your IP PBX service provider requires a port other than 5060 or 5061, enter the port numbers in the **Primary Server/Outbound Proxy Port** and **Secondary Server/Outbound Proxy Port** fields.
- To restrict or reorder the audio codecs used, modify the list in the **PBX Codecs List**. By default, the list value is *G722, G711U, G711A, G729AB*. For example, to only use only G.711, edit the list to include only *G711U, G711A*.

For specific instructions, see the configuration guide for your conferencing service provider or call control platform.

4. Expand the **Credential** section, and in the **User Credential Name** and **User Credential Password** fields, enter the SIP authentication user name and password, respectively.

If your IP PBX service provider requires a specific realm for authentication, enter it in the **Server Realm** field.

For specific instructions, see the configuration guide for your call control platform.

Related information

[Prerequisites for configuring IP PBX](#) on page 41
[Support for SIP high availability](#) on page 42
[Configuring IP PBX using the Dolby Conferencing Console](#) on page 44
[Adjusting codec preferences](#) on page 46
[Enabling Opus on Dolby Voice Huddle](#) on page 47
[Configuring voice activation detection](#) on page 48
[Configuring IP PBX for SIP URI dialing](#) on page 48
[SIP video configuration](#) on page 49

5.8.4 Configuring IP PBX using the Dolby Conferencing Console

You can use the Dolby Conferencing Console to configure IP PBX for the Dolby Voice Huddle system.

Procedure

1. Create a profile, and configure these parameters:
 - In **Features > Operation Mode**:
 - In **IP PBX Settings > Server**:

Configuration parameter	Setting
Sip.Pbx.Domain	The name of the SIP domain, if one is available.
Sip.Pbx.OutboundProxy1	The outbound proxy server address.
Sip.Pbx.OutboundProxyPort1	The outbound proxy server port number. Change the default value of 5060 if your IP PBX service provider requires a different port.
Sip.Pbx.OutboundProxy2	An alternative outbound proxy server address, if one is available.
Sip.Pbx.OutboundProxyPort2	The alternative outbound proxy server port number. Change the default value of 5060 if your IP PBX service provider requires a different port.
Sip.Pbx.SipUriScheme	Configures the URI scheme. Options are SIP and SIPS. The default is On = SIP.

Configuration parameter	Setting
Sip.Pbx.OnlyAcceptCallServerEvents	Configuration to allow SIP events from the call server only. If set to True, allows SIP events from the call server only. If set to False, allows SIP events from any address.
Sip.Pbx.CallFailoverMode	Configures call failover mode. Select one of these: <i>Active Server</i> - Call is always sent to the active server. <i>Preferred Server's Order</i> - Call is sent to the server in the order of preference.
Sip.Pbx.RegistrationTimeout	SIP registration time out. The range is from 4-59. The default is 32.
Sip.Pbx.CallFailoverTimeout	Call failover timeout. Range is 4-60. The default is 4.
Sip.Pbx.Codecs	Audio codecs list. Audio codecs used: <i>G722, G711U, G711A, G729AB</i> by default. Modify the list to specify allowable codecs or change order of preference. For example, to use only G.711, edit the list to include only <i>G711U, G711A</i> .
Sip.Pbx.InBandDTMF	Default is off. When set to <i>Off</i> out-of-band dual-tone multifrequency (DTMF) is enabled.
Sip.Pbx.EnableP2PSIPVideo	Enables Peer-to-peer SIP video calls through IP-PBX server. Default is off.
Sip.Pbx.ContentShareEnabled	Configures content share for SIP video calls is enabled or not. The default is <i>On</i> .
Sip.Pbx.URIDialingEnabled	Enables calls by URI dialing.
Sip.Pbx.URIDialingTransport	URI dialing transport. Select one of these: <i>None</i> <i>TLS</i> <i>TCP</i> <i>UDP</i>
Sip.Pbx.URIDialingProxy	URI dialing proxy server address.
Sip.Pbx.EnableURIDialingProxy	URI Dialing Proxy server to route all the SIP URI calls, ignoring URIProxy parameter.
Sip.Pbx.TLSCertNameMatchingPolicy	Matches the DNS name with the certificate name using a wildcard rule. Select: <i>Strict Wildcard Matching</i> - If the certificate name is <i>*.abc.com</i> and the DNS name is <i>1234.abc.com</i> they are matched, but DNS name <i>abc.com</i> does not. <i>Lenient Wildcard Matching</i> - Includes <i>Strict Wildcard Matching</i> and allows DNS name <i>abc.com</i> to be matched with <i>*.abc.com</i> <i>Ignore name matching</i> - No name matching at all. If the certificate is valid it is accepted. The default is <i>Strict Wildcard Matching</i>.

2. Assign the profile you created to the device pool that contains the you are configuring.

3. For each system in the device pool, set these parameters:

- In **IP PBX Settings > Account**:

Configuration parameter	Setting
Sip.Account.DisplayName	The name of the device.
Sip.Account.UserName	The extension or SIP address.
Sip.Account.DisplayNumber	The dial-in number, if different than the extension number/address.
Sip.Account.TransportType	The transport type. Select <i>AUTO</i> <i>UDP</i> <i>TCP</i> <i>TLS</i> <i>NAPTR</i> The default value is <i>AUTO</i> .

- In **IP PBX Settings > Credential:**

Configuration parameter	Setting
Sip.Credential.Name	The SIP authentication user name
Sip.Credential.Password	The SIP authentication user password
Sip.Credential.Realm	The realm for authentication, if your IP PBX service provider requires a specific realm

For specific instructions, see the configuration guide for your call control platform and the *Dolby Conferencing Console Administrator's Guide*.

Related information

- [Prerequisites for configuring IP PBX](#) on page 41
- [Support for SIP high availability](#) on page 42
- [Configuring IP PBX using the web interface](#) on page 43
- [Adjusting codec preferences](#) on page 46
- [Enabling Opus on Dolby Voice Huddle](#) on page 47
- [Configuring voice activation detection](#) on page 48
- [Configuring IP PBX for SIP URI dialing](#) on page 48
- [SIP video configuration](#) on page 49

5.8.5 Adjusting codec preferences

Dolby Voice Huddle allows you to adjust the IP PBX codec preferences using the `Sip.Pbx.Codecs` parameter.

About this task

When the device functions as a VoIP phone, the `Sip.Pbx.Codecs` parameter determines which audio codecs are used and their priority order. By default, this parameter is set to *G722*, *G711U*, *G711A*, *G729AB*. You can add or remove codecs and change the order of codecs in the list. For example, you might want to move *G729AB* to the beginning of the list to reduce VoIP traffic across the wide area network (WAN).

This parameter is not required when the system is used in **Conferencing Only Mode**.

Procedure

- From the web interface:
 - a) On the **Settings** tab, expand **IP PBX Settings**.

- b) Expand the **Server** section.
- c) In the **PBX Codec List** field, add, remove, or change the order of codecs as needed.
- From the Dolby Conferencing Console:
 - a) From the home screen, select the device pool that contains the device for which you want to edit parameters.
 - b) Select the device you want to edit.
 - c) In the left navigation bar, click the **Edit** icon.
The device configuration screen displays.
 - d) Expand **IP PBX Settings > Server**.
 - e) Drag the **PBX Codec List** to the device configuration.
The **Sip.Pbx.Codecs** parameter appears in the device configuration.
 - f) In the **Sip.Pbx.Codecs** field, add, remove, or change the order of codecs as needed.

Related information

[Prerequisites for configuring IP PBX](#) on page 41
[Support for SIP high availability](#) on page 42
[Configuring IP PBX using the web interface](#) on page 43
[Configuring IP PBX using the Dolby Conferencing Console](#) on page 44
[Enabling Opus on Dolby Voice Huddle](#) on page 47
[Configuring voice activation detection](#) on page 48
[Configuring IP PBX for SIP URI dialing](#) on page 48
[SIP video configuration](#) on page 49

5.8.6 Enabling Opus on Dolby Voice Huddle

You can enable Opus on Dolby Voice Huddle for WebRTC calls. Consult your conferencing service provider confirm if Opus must be turned on.

You can enable Opus by setting parameters or through the web UI.

Set the configuration parameters:

To enable Opus for Dolby Voice Huddle, set the `Dvms.Audio.EnableOpus` parameter to true. By default, this parameter is set to false.

Use the web UI :

1. Navigate to **Settings > Dolby Voice Conferencing Server > Audio**
2. Toggle the **Enable Opus** button.

Related information

[Prerequisites for configuring IP PBX](#) on page 41
[Support for SIP high availability](#) on page 42
[Configuring IP PBX using the web interface](#) on page 43
[Configuring IP PBX using the Dolby Conferencing Console](#) on page 44
[Adjusting codec preferences](#) on page 46
[Configuring voice activation detection](#) on page 48
[Configuring IP PBX for SIP URI dialing](#) on page 48
[SIP video configuration](#) on page 49

5.8.7 Configuring voice activation detection

The Dolby Voice Huddle by default signals G.729AB (G.729A with Annex B) to its peer. If your IP PBX configuration requires the device to signal G.729A, you can set the `Sip.Vad.SignalAnnexB` configuration parameter to `false` to override the default behavior.

About this task

The Dolby Voice Huddle supports the G.729AB codec, but it also allows Annex B in its Session Description Protocol (SDP) to be turned off.

You can perform this procedure from the web interface or from Dolby Conferencing Console.

Procedure

1. From the web interface:
 - a) From the **Settings** tab, expand the **IP PBX Settings** section, and then expand the **VAD** section.
 - b) Turn off `Signal G729 Annex B`.
This option controls the `Sip.Vad.SignalAnnexB` parameter.
2. Alternatively, from Dolby Conferencing Console:
 - a) From the home screen, select the device pool that contains the device for which you want to edit parameters.
 - b) Select the device you want to edit.
 - c) In the left navigation bar, click the **Edit** icon.
The device configuration screen displays.
 - d) Navigate to **IP PBX Settings > VAD**.
 - e) Drag `Signal G729 Annex B` to the device configuration in the upper right-hand side of the screen.
The `Sip.Vad.SignalAnnexB` parameter appears in the device configuration.
3. Turn the `SIP.Vad.SignalAnnexB` parameter off.

Results

The device now signals peers using G.729A.

Related information

- [Prerequisites for configuring IP PBX](#) on page 41
- [Support for SIP high availability](#) on page 42
- [Configuring IP PBX using the web interface](#) on page 43
- [Configuring IP PBX using the Dolby Conferencing Console](#) on page 44
- [Adjusting codec preferences](#) on page 46
- [Enabling Opus on Dolby Voice Huddle](#) on page 47
- [Configuring IP PBX for SIP URI dialing](#) on page 48
- [SIP video configuration](#) on page 49

5.8.8 Configuring IP PBX for SIP URI dialing

You can use the web interface or the Dolby Conferencing Console to configure IP PBX for dialing using a SIP URI.

About this task

By default, SIP URI dialing is enabled and the configuration parameter is set to **TRUE**.

Procedure

- To disable SIP URI dialing from the web interface:
 - a) On the **Settings** tab, expand **IP PBX Settings**.

- b) Expand the **Server** section.
- c) Turn off **Enable SIP URI Dialing**.
- To disable SIP URI dialing from the Dolby Conferencing Console:
 - a) From the home screen, select the device pool that contains the device for which you want to edit parameters.
 - b) Select the device you want to edit.
 - c) In the left navigation bar, click the **Edit** icon.
The device configuration screen displays.
 - d) Expand **IP PBX Settings > Server**.
 - e) Drag **Enable SIP URI Dialing** to the device configuration.
The `Sip.Pbx.URIDialingEnabled` parameter appears in the device configuration.
 - f) Set **Sip.Pbx.URIDialingEnabled** to **FALSE**.

Related information

[Prerequisites for configuring IP PBX](#) on page 41
[Support for SIP high availability](#) on page 42
[Configuring IP PBX using the web interface](#) on page 43
[Configuring IP PBX using the Dolby Conferencing Console](#) on page 44
[Adjusting codec preferences](#) on page 46
[Enabling Opus on Dolby Voice Huddle](#) on page 47
[Configuring voice activation detection](#) on page 48
[SIP video configuration](#) on page 49

Making a call using SIP URI dialing

You can make a call using the SIP URI dialer after configuration through the web interface or the Dolby Conferencing Console. SIP URI dialing works for unregistered and unauthenticated SIP lines.

About this task

The Dolby Voice Huddle keyboard field supports a maximum of 128 characters.

Procedure

1. On the Dolby Voice Huddle UI, tap the **URI** button.
The onscreen keyboard displays with **sip:**.
2. Enter the URI, and tap the **Call** button to join the call.

5.8.9 SIP video configuration

Dolby Voice Huddle allows you to dial into third-party video conferences using SIP URI.

On Dolby Voice Huddle, SIP video calling is enabled by default. Dialing into video conferences using SIP URI does not require SIP registration. The default video codec used for URI-based video calls is H.264.

You can make video calls using UDP, TCP or TLS transport. The default is TLS. To make a call using TCP transport, the URI format is `user@domain.com;transport=tcp`. To make a call using TLS transport, the URI format is `user@domain.com;transport=tls`.

You also can make a SIP video call using only the host name or IP address.

If a video call needs to pass through a proxy server, you have to set the `Sip.Pbx.URIDialingProxy` parameter to specify the proxy server and set `uriproxy=1` to the URI. You can set this parameter in the Dolby Voice Huddle web UI configuration page by navigating to **IP PBX Settings > Server**.

If the `uriproxy=1` parameter is set, but the `Sip.Pbx.URIDialingProxy` parameter is not set, Dolby Voice Huddle will use the IP PBX configuration to make calls.

On video calls, you can:

- Disable the camera
- Use Room mode to capture the entire room

For screen sharing, Dolby Voice Huddle supports full 1080p high definition (HD) content share.

Dolby Voice Huddle provides a lock icon to identify a secure video connection. If the lock icon is white and locked, the connection is secure, and audio and video are transmitted using SRTP. If it is red and unlocked, it is insecure, and audio and video are transmitted using RTP.

Related information

- [Prerequisites for configuring IP PBX](#) on page 41
- [Support for SIP high availability](#) on page 42
- [Configuring IP PBX using the web interface](#) on page 43
- [Configuring IP PBX using the Dolby Conferencing Console](#) on page 44
- [Adjusting codec preferences](#) on page 46
- [Enabling Opus on Dolby Voice Huddle](#) on page 47
- [Configuring voice activation detection](#) on page 48
- [Configuring IP PBX for SIP URI dialing](#) on page 48

Making a SIP video call

You can dial into third-party video conferences using SIP URI.

About this task

You can make video calls using UDP, TCP or TLS transport. The default is TLS. To make a call using TCP transport, the URI format is `user@domain.com;transport=tcp`. To make a call using TLS transport, the URI format is `user@domain.com;transport=tls`. The Dolby Voice Huddle keyboard field supports a maximum of 128 characters.

You can use the call log to view what type of calls were made on Dolby Voice Huddle by tapping the call log button. If it was a video call, a video icon displays.

Procedure

1. On the Dolby Voice Huddle UI, tap the call button.
2. On the keypad screen, tap the keyboard icon .
The onscreen keyboard displays.
3. Enter the URI, and tap the **Call** button to join the call.

 **Note:** You must add the prefix SIP: before the URI. Otherwise, it will be treated as an IP- PBX call.

`sip:6505551212.com`

5.9 Configuring preferences

You can set up these preferences for your users, or you can allow users to set them themselves.

5.9.1 Disabling the recent-calls list

The Dolby Voice Huddle system can gather and display a list of the 100 most recently missed, placed, and received calls. This list is enabled by default.

About this task

The recent-calls list contains call information such as remote-party identification, time and date of the call, and call duration. The list is uploaded from the system to the provisioning server after each call is complete.

The list can be used to redial previous outgoing calls, return missed calls, and save contact information to the local contact directory. The list can be cleared manually by individual users. The list is retained when the system reboots or when the system software is upgraded.

Procedure

- Disable the recent-calls list with the web interface:
 - a) Access the web interface.
 - b) In the **Settings** tab, expand **Features**, and turn off **Enable Recent Calls**.
- Disable the recent-calls list by setting a configuration parameter.
 - a) Open the diagnostic shell.
 - b) Enter this command:

```
configure Features.RecentCallsEnabled false
```

Related information

[Enabling user feedback](#) on page 51

[Configuring call forwarding](#) on page 51

[Enabling automatic answer](#) on page 52

[Enabling do not disturb](#) on page 53

[Setting the language](#) on page 54

[Setting the time and date display](#) on page 54

5.9.2 Enabling user feedback

You can set up the Dolby Voice Huddle system to solicit feedback from users about their experience with the phone, from a simple thumbs-up to requesting a call to provide more detailed feedback. By default, this feature is enabled.

Procedure

- Set these configuration parameters to enable user feedback.

Configuration parameter name	Configuration parameter description
Features.UserFeedback.Enabled	Enable or disable user feedback.
Features.UserFeedback.HotLine	Call user feedback hotline.

Related information

[Disabling the recent-calls list](#) on page 50

[Configuring call forwarding](#) on page 51

[Enabling automatic answer](#) on page 52

[Enabling do not disturb](#) on page 53

[Setting the language](#) on page 54

[Setting the time and date display](#) on page 54

5.9.3 Configuring call forwarding

The system provides a flexible call-forwarding feature that enables you to forward incoming calls to another party.

About this task

There are three ways to apply call forwarding:

- To all calls
- When the system is busy
- When a call is not answered after 40 seconds

There are several ways to configure call forwarding:

Procedure

- Use the web UI:
 - a) Navigate to **Settings > User Preferences**.
 - b) Enter the phone number in the appropriate forwarding options.
- Use the controller UI:
 - a) Navigate to the Dolby Voice Huddle settings menu.
This may be different for each service provider.
 - b) Tap **Telephony**.
 - c) Select **Call Forward All Number**, **Call Forward No Answer Number**, or **Call Forward Busy Number**.
 - d) Enter the phone number where you want the calls forwarded.
- Set these configuration parameters:

Configuration parameter name	Configuration parameter description
Preferences.ForwardAllNumber	Enter the number where all calls are forwarded.
Preferences.ForwardBusyNumber	Enter the number where the call is forwarded if the system is busy.
Preferences.ForwardNoAnswerNumber	Enter the number where the call is forwarded if the user does not answer.

To disable any of these call-forwarding options, specify a null string:

```
configure Preferences.ForwardAllNumber ""
```

Related information

[Disabling the recent-calls list](#) on page 50

[Enabling user feedback](#) on page 51

[Enabling automatic answer](#) on page 52

[Enabling do not disturb](#) on page 53

[Setting the language](#) on page 54

[Setting the time and date display](#) on page 54

5.9.4 Enabling automatic answer

In certain circumstances, you may be unable to physically answer an incoming call. This feature allows you to configure the system to answer incoming calls automatically after the first ring. The call is answered with the microphone muted.

About this task

There are several ways to configure automatic answer:

Procedure

- Set the configuration parameter:

Configuration parameter name	Configuration parameter description
Preferences.AutoAnswerEnabled	Enable or disable automatic answer. Set to true or false.

- Use the webUI:
 - a) Go to **Settings > User Preferences**.
 - b) Select the automatic answer **Off/On** option.
- Use the controller UI:
 - a) Navigate to the Dolby Voice Huddle settings menu.
This may be different for each service provider.
 - b) Tap **Telephony > Auto-Answer**.
 - c) Toggle the **Auto-Answer** button.

Related information

[Disabling the recent-calls list](#) on page 50
[Enabling user feedback](#) on page 51
[Configuring call forwarding](#) on page 51
[Enabling do not disturb](#) on page 53
[Setting the language](#) on page 54
[Setting the time and date display](#) on page 54

5.9.5 Enabling do not disturb

You use the do not disturb feature to temporarily stop incoming calls and turn off audio alerts.

About this task

While do not disturb is turned on, incoming calls are sent a busy signal and are logged as missed.

There are several ways to configure the do not disturb feature.

Procedure

- Set the configuration parameter:

Configuration parameter name	Configuration parameter description
Preferences.DndEnabled	Enable or disable do not disturb behavior. Set to true or false.

- Use the web UI:
 - a) Go to **Settings > User Preferences**.
 - b) Select the Do Not Disturb **Off/On** option.
- Use the controller UI:
 - a) Navigate to the Dolby Voice Huddle settings menu. This may be different for each service provider.
 - b) Tap **Telephony > Do Not Disturb**.
 - c) Toggle the **Do Not Disturb** button.

Related information

[Disabling the recent-calls list](#) on page 50
[Enabling user feedback](#) on page 51
[Configuring call forwarding](#) on page 51
[Enabling automatic answer](#) on page 52
[Setting the language](#) on page 54

[Setting the time and date display](#) on page 54

5.9.6 Setting the language

You can change the language displayed on the controller UI. By default, English is shown.

About this task

There are several ways to set the language.

Procedure

- Set the configuration parameter:

Configuration parameter name	Configuration parameter description
Preferences.Localization.Language	Possible values include: - English - French - German - Italian - Spanish - Portugal Portuguese - Brazilian Portuguese - Simplified Chinese - Traditional Chinese - Japanese - Korean - Russian - Czech - Hebrew - Arabic

- Use the web interface:
 - Navigate to **Settings > User Preferences > Localization**.
 - Select the language from the **Language** list.
- Use the controller UI:
 - Access the Dolby Voice Huddle setting menu.
 - Tap **Preferences > Language**
 - Select the language from the list.

Related information

[Disabling the recent-calls list](#) on page 50

[Enabling user feedback](#) on page 51

[Configuring call forwarding](#) on page 51

[Enabling automatic answer](#) on page 52

[Enabling do not disturb](#) on page 53

[Setting the time and date display](#) on page 54

5.9.7 Setting the time and date display

A clock and calendar are enabled by default. You can display the time and date for your time zone in several formats. The system will synchronize with the configured time server.

About this task

The available date formats may change, depending on the language selection.

There are several ways to set the time and date display.

Procedure

- Set the configuration parameters:

Configuration parameter name	Configuration parameter description
Preferences.Localization.Timezone	Time zone. Permitted values are: • Pacific/Kwajalein • Pacific/Pitcairn • Pacific/Samoa • Pacific/Honolulu • America/Anchorage • America/Los_Angeles • America/Denver • America/Mazatlan • America/Phoenix • America/Chicago • America/Guatemala • America/Mexico_City • America/Bogota • America/New_York • America/Asuncion • America/Caracas • America/Campo_Grande • America/Santiago • America/Sao_Paulo • America/St_Johns • America/Argentina/Buenos_Aires • Atlantic/South_Georgia • Atlantic/Azores • UTC • Europe/London • Africa/Lagos • Europe/Paris • Africa/Johannesburg • Asia/Jerusalem • Asia/Baghdad • Asia/Tehran • Asia/Muscat • Asia/Kabul • Asia/Karachi • Asia/Kolkata • Asia/Kathmandu • Asia/Dhaka • Asia/Rangoon • Asia/Bangkok • Asia/Singapore • Asia/Tokyo • Australia/Adelaide • Australia/Darwin • Australia/Lord_Howe • Australia/Sydney • Asia/Magadan • Pacific/Auckland • Pacific/Tongatapu
Preferences.Localization.TimeFormat	Time format. Permitted values are: • <i>h:mm ap</i> (For 12-hour format. Enclose in quotes when specifying this value in the configure command.) • <i>h:mm</i> (For 24-hour format)
Preferences.Localization.DateFormat	Date format. Permitted values are:

- Use the web interface:
Go to **Settings > User Preferences > Localization**, and adjust the time and date parameters.
- Use the controller UI:
 - To change the time zone, access the Dolby Voice Huddle setting menu. Tap **Preferences > Time Zone**, and select the desired time zone from the list.
 - To change the time format, access the Dolby Voice Huddle setting menu. Tap **Preferences > Time Format**, and select the desired format from the list.
 - To change the date format, access the Dolby Voice Huddle setting menu. Tap **Preferences > Date Format**, and select the desired format from the list.

Related information

[Disabling the recent-calls list](#) on page 50

[Enabling user feedback](#) on page 51

[Configuring call forwarding](#) on page 51

[Enabling automatic answer](#) on page 52

[Enabling do not disturb](#) on page 53

[Setting the language](#) on page 54

5.10 Configuring contact directories

For making IP PBX calls, Dolby Voice Huddle enables users to access a corporate directory and to create a local directory.

5.10.1 Corporate directory

Dolby Voice Huddle can connect to a directory server that supports the LDAP version 3.

You can configure LDAP directory parameters to make your corporate directory available. After it is set up, users can search the corporate directory.

If you are required to access the LDAP server in a secure manner, you must download the appropriate CA certificate to the system.

Only Microsoft Active Directory 2003 SP2 or later is supported at this time. Microsoft Active Directory supports server-side sorting.

Tip:

Consult your LDAP administrator when changing any corporate directory configuration information. For more information on LDAP attributes, see IETF RFC 4510.

Related information

[Local directory](#) on page 57

[Configuring the corporate directory](#) on page 58

[Creating a local contact directory from the provisioning server](#) on page 59

[Resetting the directory](#) on page 59

[Resetting call logs](#) on page 59

5.10.2 Local directory

A local directory on the controller can hold up to 1,000 contacts.

You can enter contacts to a local contact directory manually or save recent calls and corporate directory search results.

 **Note:** For saved entries to work, you must format them in a manner that is consistent with your call-control platform. For example, you may need to include the number 9 or a plus sign (+) at the beginning of the entry.

A local directory can also be downloaded from the provisioning server. Users can then manually add entries to the downloaded directory.

Related information

[Corporate directory](#) on page 57

[Configuring the corporate directory](#) on page 58

[Creating a local contact directory from the provisioning server](#) on page 59

[Resetting the directory](#) on page 59

[Resetting call logs](#) on page 59

5.10.3 Configuring the corporate directory

Use the web interface or set configuration parameters to configure an LDAP corporate directory.

About this task

You need this information to connect the system to the LDAP directory:

- The IP address or host name of the LDAP server
- The log-in name and password for the LDAP server
- The base domain name for the search of the LDAP server

 **Note:** Certain parameters are set by default (for example, search filter, scope, time-out, limit, and attributes).

Procedure

1. Do one of the following :
 - a) Specify LDAP parameters with the web interface:
Navigate to **Settings > Corporate Directory > LDAP**.
 - b) Set LDAP configuration parameters, either with the diagnostic shell or from Dolby Conferencing Console.

Parameter	Description
Directory.Ldap.URIList	Server URI. Use a comma-separated list to add more than one server. For example: ldap://10.101.21.5,ldap://10.120.21.6
Directory.Ldap.LoginName	Log-in name
directory.Ldap.Password	Password
Directory.Ldap.BaseDN	Base domain name. For example, dc=example, dc=com
Directory.Ldap.SecureConnection	Set to <i>true</i> or <i>false</i> , depending on the security settings on the LDAP server.

2. Reboot the system.

Results

After the reboot is complete, users can search for contacts in the LDAP server.

Related information

[Corporate directory](#) on page 57

[Local directory](#) on page 57

[Creating a local contact directory from the provisioning server](#) on page 59

[Resetting the directory](#) on page 59

[Resetting call logs](#) on page 59

5.10.4 Creating a local contact directory from the provisioning server

On the provisioning server, you can create a local contact directory for users.

When the system reboots, the local contact directory is downloaded from the provisioning server. The user can then modify the downloaded directory by adding contacts, either manually or by saving recent calls or corporate-directory search results.

See the *Dolby Conferencing Console Administrator's Guide* for details.

 **Note:** Dolby Conferencing Console version 1.2 and later allows you to provision a local contact directory.

Related information

[Corporate directory](#) on page 57

[Local directory](#) on page 57

[Configuring the corporate directory](#) on page 58

[Resetting the directory](#) on page 59

[Resetting call logs](#) on page 59

5.10.5 Resetting the directory

Use the Dolby Voice Huddle Controller to remove all the entries in the local directory.

Procedure

1. Navigate to the Dolby Voice Huddle settings menu.
This may be different for each service provider.
2. Tap **Telephony > Reset Directory**.
The **Reset Directory** screen appears.
3. Click on the confirm button to remove all the entries in the local directory.

Related information

[Corporate directory](#) on page 57

[Local directory](#) on page 57

[Configuring the corporate directory](#) on page 58

[Creating a local contact directory from the provisioning server](#) on page 59

[Resetting call logs](#) on page 59

5.10.6 Resetting call logs

Use the Dolby Voice Huddle Controller to reset the call log and clear the list of recent calls.

Procedure

1. Navigate to the Dolby Voice Huddle settings menu.
This may be different for each service provider.
2. Tap **Telephony > Reset Call Log**.

The **Reset Call Log** screen appears.

3. Click on the confirm button to clear the list of recent calls.

Related information

[Corporate directory](#) on page 57

[Local directory](#) on page 57

[Configuring the corporate directory](#) on page 58

[Creating a local contact directory from the provisioning server](#) on page 59

[Resetting the directory](#) on page 59

5.11 Configuring the time server

The Dolby Voice Huddle system uses a time server to synchronize the clocks on all of the on the network. If your network has a local NTP server, we highly recommend that you configure it.

About this task

The Dolby Voice Huddle system uses DHCP option 42 to configure the time server. If the system cannot configure the time server using DHCP option 42, and the system is connected to an IP PBX, the IP PBX supplies the time, and the NTP status is shown as **Synchronized with SIP** on the Dolby Voice Huddle UI and web interface.

If you perform a factory reset, and DHCP option 42 is not received and the NTP servers are not provisioned manually, the time servers are set by default to *time.nist.gov* as the primary NTP server and *pool.ntp.org* as the secondary NTP server.

 **Note:** The network must have a DNS server for the system to resolve the host names and connect to the default NTP servers. In addition, your firewall policy must permit access to these servers.

You can set and check the time server parameters in several ways:

Procedure

- Set the configuration parameters:

Configuration parameter name	Configuration parameter description
Network.Ntp.AutoDiscover	Enables NTP auto discovery (default true). When set to false, DHCP option 42 is ignored and you can manually set NTP server addresses with specified IP addresses or hostnames.
Network.Ntp.Server1	Sets the primary NTP server IP address or hostname
Network.Ntp.Server2	Sets the secondary NTP server IP address or hostname

- Use the web interface:
 - a) Go to **Settings > Network > NTP** and enter the parameters.
- Use the controller UI:
 - a) To review the time server status, access the Dolby Voice Huddle settings menu.
 - b) Tap **Administrative Settings**.
 - c) Enter the admin password.
 - d) Tap **Network Configuration > Network Time Server > Primary NTP Server Address**.

5.12 Calibrating audio

You can calibrate the Dolby Voice Huddle audio using the Dolby Voice Huddle Controller.

About this task

Dolby Voice Huddle generates audio output from the display speakers and listens to the audio using the microphone, and uses the input audio to self-calibrate to optimize audio experience/quality.

For maximum sound quality, it is best to use high quality speakers and displays with low latency and minimum sound processing features (less sound processing features mean less delay).

Procedure

1. Disable all the audio presets from your display controller and have the volume on to adjust the level.
2. Navigate to the Dolby Voice Huddle settings menu.
This may be different for each service provider.
3. Tap **General Settings > Audio Calibration**.
4. Tap **Launch**.
5. Tap **Play Tone**.
6. Set the volume to a comfortable level and tap **Done**.
7. Tap **Start Test** to begin the calibration test.
8. Tap **Finish** once the calibration is complete.

5.13 Disabling the reboot prompt

The reboot prompt is disabled by default, prompting you to reboot due to configuration changes. If you take no action in 20 seconds—for example, cancel the reboot—the system reboots automatically.

About this task

This procedure disables both the reboot prompt.

You can use either configuration parameters or the web interface to disable the reboot prompt.

Procedure

- Set the configuration parameter:

Configuration parameter name	Configuration parameter description
Provisioning.CfgChange. UserAcknowledgementEnabled	Enables/disables the reboot. Set to false.

- Use the web interface:
 - a) Go to **Settings > Provisioning > On Configuration Change**.
 - b) Set **User acknowledgment enabled** to **Off**.

6

Administering Dolby Voice Huddle

Administrative tasks include viewing system status, viewing and managing event logs, viewing call statistics, running diagnostics, and more.

- [Changing the administrative password](#)
- [Accessing the Dolby Voice Huddle web interface](#)
- [Administering with the diagnostic shell](#)
- [Viewing the status menu](#)
- [Setting the Dolby Voice Huddle operational mode](#)
- [Viewing and managing event logs](#)
- [Initiating Reprovisioning](#)
- [Enabling power saving mode](#)
- [Changing network settings](#)
- [Enabling Privacy block](#)
- [Call and audio statistics](#)
- [Enabling WebRTC tracing](#)
- [Rebooting Dolby Voice Huddle](#)
- [Resetting to factory defaults](#)
- [Updating the Dolby Voice Huddle software](#)
- [Configuring HDMI input audio](#)
- [Finding the system serial number and software version](#)
- [Finding the serial number and software version](#)

6.1 Changing the administrative password

The Dolby Voice Huddle administrative settings are protected with an administrative password. The same password is used to log in to the web interface and diagnostic shell.

About this task

The default administrative user name is *admin* and the default password is *1739*. Change the administrative password from the default value as soon as possible. If you leave the administrative password unchanged or use weak passwords, the system is vulnerable to various network attacks.

 **Note:** If you changed and then subsequently forgot the administrative password, reset the system to factory defaults to restore the default password of *1739*. If you are not already logged in, you must use the fail-safe method to reset the system.

Use one of these methods to change the administrative password.

 **Note:** We recommend that you do not enter passwords in configuration files, as it poses a security risk.

Procedure

- Use the web interface.
 - a) Access the Dolby Voice Huddle web interface, and log in with the current administrative user name and password.
 - b) Go to **Settings > Authentication > Local Account**, enter a new administrative password, and click **Save**.
- Use the controller UI.
 - a) Navigate to the Dolby Voice Huddle settings menu.
This may be different for each service provider.
 - b) Tap **Administrative Settings > Change Password**.
 - c) Enter and confirm the new password.
 - d) Tap the confirm button to save the new password.
- Use the diagnostic shell and the **configure** command.
 - a) Open the diagnostic shell, and log in with the current administrative user name and password.
 - b) Enter this command:

```
configure Authentication.Local.Admin newPassword
```

6.2 Accessing the Dolby Voice Huddle web interface

Use the web interface to view and change Dolby Voice Huddle settings.

About this task

You can access the Dolby Voice Huddle web interface using any of these web browsers:

- Microsoft Internet Explorer 10.0.x or later on Windows-based computers (backward compatibility not supported)
- Google Chrome 31.0.x or later
- Apple Safari 7.0.2.x

 **Note:** The Dolby Voice Huddle web interface has been optimized for modern browsers. The best experience is obtained using one of the supported browsers. Versions 8 and 9 of Internet Explorer work in a degraded fashion; support for these browsers is deprecated and will be removed in a future release.

By default, access to the web interface is enabled. You can disable access only through the configuration files by changing `Network.Services.WebServerMode`.

Procedure

1. Determine the system IP address:
 - a) Access the Dolby Voice Huddle **Settings** screen.
Ask your conferencing service provider for instructions.
2. Open one of the supported web browsers.
3. Enter `http://` in the browser address bar.
The log-in page displays.

 **Note:** If the browser displays a certificate-related error message, such as **Site is not secure** or **Your connection is not private**, then take the required steps to bypass the error page and proceed to the log-in page.

4. Log in as the administrator. (Default credentials are `admin` and `1739`.)
The web interface main page displays. Expand the headings and hover the mouse over a parameter name to see a parameter description, along with the name of the corresponding configuration parameter.

 **Note:** We recommend that you change the default administrative password.

6.3 Administering with the diagnostic shell

The diagnostic shell provides a command-line interface for administering Dolby Voice Huddle.

6.3.1 Diagnostic shell

The diagnostic shell provides a command-line interface for you to query the system state, run diagnostics, and configure selected parameters.

Only one instance of the shell service is supported. After a connection to the shell service is established, subsequent requests are denied until the current session is terminated. The session times out after a period of inactivity. The time-out period is set by the `Authentication.Local.SessionTimeout` configuration parameter. The shell service starts when the system boots up, but it is accessible remotely only if the configuration parameter `Network.Services.SSHEnabled` is enabled, which is the default. You must enter the administrative user name and password to use the diagnostic shell.

Shell commands include localized command help, function calls, diagnostic tests, commands to view and set configuration parameters, and more. A list of commands is available in the **Help** section of the web interface.

Figure 3: Diagnostic shell command sample

```

DolbyCli>show sysload
Overall CPU Load   : 32%
Services CPU Load  : 2%
GUI CPU Load       : 0%
Overall Memory Used: 1632888 kB
Services Memory Used:32536 kB
GUI Memory Used    : 157456 kB
DolbyCli>show memory
Memory Usage
Total 4055339 kB
Free 2383261 kB
Used 1672078 kB
Used% 41.231505
DolbyCli>configure Logging.Level.Sip INFO
DolbyCli>configure Logging.Level.Web INFO
DolbyCli>

```

Tip: The shell provides help and shortcuts:

- Type *help* to see a list of commands.
- Press Up-arrow to view previous commands. Press Enter to submit.
- Press Tab to complete partially typed commands or to prompt for additional command information, such as command parameters. For example, if you enter *system*, and then press Tab, the possible system commands are listed and the command is repeated on the command line for you to finish command entry. If you enter *system traceroute* and press Tab, the system indicates that the command parameter is the IP address of the test host.

Related information

[Opening the diagnostic shell](#) on page 65

6.3.2 Opening the diagnostic shell

You access the diagnostic shell remotely with the SSH protocol or locally with an external keyboard.

- Open the diagnostic shell remotely:
 1. Determine the system IP address:
 - a. Access the Dolby Voice Huddle **Settings** screen. This may be different for each service provider.
 - b. Tap this sequence:
About > System.

The IP address displays.
 2. Open your preferred SSH client application.
 3. Connect to the system IP address using SSH (port 22).
 4. Log in with the administrative user name and password. (Default credentials are *admin* and *1739*.)
 5. Enter commands, and then type *exit* to exit the shell.

Related information

[Diagnostic shell](#) on page 64

6.4 Viewing the status menu

The status menu on the Dolby Voice Huddle Controller UI displays status information on the Dolby Voice Huddle components, the provisioning server, the network, and more.

Procedure

1. Navigate to the Dolby Voice Huddle settings menu. This may be different for each service provider.
2. Tap **About > Status**.

6.5 Setting the Dolby Voice Huddle operational mode

Procedure

- Use the web interface:
 - a) Access the Dolby Voice Huddle web interface.
 - b) Go to **Settings > Features** and, for **Operation Mode**, select **Conferencing Only Mode** or **Dual Mode**.
- Use the diagnostic shell:
 - a) Open the diagnostic shell.
 - b) Use the **configure** command to set the `Features.OperationMode` configuration parameter to one of the following values:
 - DUALMODE
 - CONFONLYMODE

Example: `configure Features.OperationMode DUALMODE`

6.6 Viewing and managing event logs

The Dolby Voice Huddle system has a sophisticated event-logging system for diagnostics and troubleshooting.

6.6.1 Event logging

The Dolby Voice Huddle system generates event-log entries on a continuous basis, both for ordinary operation events and for error conditions. The event logs provide an essential troubleshooting tool. Dolby Customer Support or your conferencing service provider might ask you to share logs during problem resolution.

Log files are stored in a flash file system on the Dolby Voice Huddle system. You can use the web interface to download the logs directly from the device.

If you are using automatic provisioning with Dolby Conferencing Console, Dolby Voice Huddle automatically uploads log entries to Dolby Conferencing Console, and you view the logs with the Dolby Conferencing Console web interface.

Dolby Voice Huddle supports the `syslog` protocol. You can configure the system to automatically upload logs to a `syslog` receiver.

For information about viewing logs on Dolby Conferencing Console, see the .

Related information

[Log levels](#) on page 67

[Audio logging](#) on page 67

[syslog logging](#) on page 68

[Enabling HDMI logging](#) on page 69

[Viewing log files](#) on page 69

6.6.2 Log levels

Each of the components of the Dolby Voice Huddle software is capable of logging events of different severity. This allows you to capture low-severity events in one component, and only high-severity events for other components.

Log-level configuration parameters enable you to indicate how much logging is done for each software component. The range of log levels, in order of detail and verbosity from lowest to highest, includes NONE, EMERGENCY, ALERT, CRITICAL, ERROR, WARNING, NOTICE, INFO, DEBUG, and DEBUG1. (DEBUG1 contains maximum verbose trace information.) By default, all log-level parameters are set to INFO.

 **Note:** You should not modify the default log levels unless directed to by Dolby Customer Support. Inappropriate logging levels can cause performance issues.

In the log files, messages are prefixed by timestamp and log level.

- `debug`: Indicates verbose traces generated only when log level is set to DEBUG
- `info`: Indicates normal controller behavior
- `notice`: Indicates minor issues or failures, but could be normal behavior
- `warn`: Indicates that an error may occur if action is not taken
- `err`: Indicates that a failure or loss of functionality occurred
- `crit`: Indicates that an unrecoverable failure of a semi-important component occurred
- `alert`: Indicates that an unrecoverable failure of an important component occurred
- `emerg`: Indicates that a complete unrecoverable system failure occurred

The `warn`, `err`, `crit`, `alert`, and `emerg` messages indicate problems of increasing severity.

A full description of all log-level parameters can be found under the **Help** section of the web interface.

To change the logging level of the SIP communication module to the DEBUG level, set `Logging.Level.Sip` to DEBUG. You can also make this change in the **Logging** section of the web interface.

 **Note:** If you set logging to a more verbose level for testing or troubleshooting, then when testing or troubleshooting is complete, remember to return logging levels to their default values.

Related information

[Event logging](#) on page 66

[Audio logging](#) on page 67

[syslog logging](#) on page 68

[Enabling HDMI logging](#) on page 69

[Viewing log files](#) on page 69

6.6.3 Audio logging

For the audio module of the Dolby Voice Huddle software, you can enable logging through configuration files or the web interface.

 **Note:** Turn on audio logging only if requested by Dolby Customer Support.

Audio logging tracks these audio characteristics:

- Echo reference input

- Echo reference output
- Jitter buffer input
- Jitter buffer output

A full description of all audio logging configuration parameters can be found in the **Help** section of the web user interface.

By default, all audio logging is disabled. If you want the audio log files written to the provisioning server, you need to set the audio logging mode, `Logging.Audio.Mode`. The supported modes are `DISABLED`, `STATISTICS`, and `DIAGNOSTIC`.

If the audio logging mode is set to `STATISTICS` or `DIAGNOSTIC`:

- A call statistics log containing per-second statistical information (including jitter buffer and packet loss) is uploaded to the provisioning server. For example, the file `D0025454-call_stats_b502bb32-2bdd-49da-8ae7-0d89ff295a01.json` is uploaded from the device numbered D0025454.
- An audio dump of preprocessing statistics that contains metadata about the preprocessing algorithm is uploaded to the provisioning server. For example, the file `D0025454-vsv_stats_20140529163458.bin` is uploaded from the device numbered D0025454.

If the audio logging mode is set to `DIAGNOSTIC`, the recorded diagnostic data is uploaded to the provisioning server. For example, the file `D0025454-audiolog_20140529141935.gz` is uploaded from the device numbered D0025454.

 **Note:** The `STATISTICS` audio logging mode may impact the end of a call (when data is uploaded to the provisioning server). The `DIAGNOSTIC` audio logging mode impacts the entire call, adding approximately 10–15% overhead.

Related information

[Event logging](#) on page 66

[Log levels](#) on page 67

[syslog logging](#) on page 68

[Enabling HDMI logging](#) on page 69

[Viewing log files](#) on page 69

6.6.4 syslog logging

`syslog` is a standard for forwarding log messages in an IP network. You can configure the Dolby Voice Huddle system to ship log entries to a `syslog` receiver.

Because `syslog` is supported by a wide variety of devices and receivers, `syslog` can be used to integrate log data from many different types of systems into a central repository.

The `syslog` messages contain much of the same information as the Dolby Voice Huddle logs, with the exception of the events during the early stage of the system initialization (as the device is not yet configured to send `syslog` messages). `syslog` data is sent in clear text.

By default, `syslog` is disabled. (`Logging.RemoteSysLog.Enable` is `false`.) To use `syslog`, you must set `Logging.RemoteSysLog.Enable` to `true` and set the `syslog` server address (`Logging.RemoteSysLog.Address`). Messages relating to developer logs and intermodule communication are sent to the `DEBUG` level. Messages pertaining to user action (if logged) and module initialization (if any) are available in the `INFO` level. Error and warning messages are logged in `ERROR` and `WARNING`, respectively.

For more information on the `syslog` protocol, see RFC 3164.

Related information

[Event logging](#) on page 66

[Log levels](#) on page 67

[Audio logging](#) on page 67

[Enabling HDMI logging](#) on page 69

[Viewing log files](#) on page 69

6.6.5 Enabling HDMI logging

You can enable HDMI logging on the Dolby Voice Huddle HDMI-input drivers.

About this task

HDMI logging can be used to troubleshoot issues with HDMI sources. The following parameters are available:

- **Enable HDMI Input Detection logs:** Provides information on how the HDMI source connected to the system is detected and information about the incoming signal. The logs can be used for detecting HDMI source connection and disconnection problems.
- **Enable HDMI Capture logs:** Provides information on the capture sequence after an HDMI source has been successfully connected and detected. When an application requests the HDMI-input stream, the subsequent reactions in the driver are logged. The logs can be used for detecting when a source is detected properly but the displayed HDMI image is corrupted.

Procedure

1. Access the Dolby Voice Huddle web interface.
2. In the **Settings** tab, expand **Logging > HDMI Input**.
3. Toggle the desired HDMI logs.

Related information

[Event logging](#) on page 66

[Log levels](#) on page 67

[Audio logging](#) on page 67

[syslog logging](#) on page 68

[Viewing log files](#) on page 69

6.6.6 Viewing log files

You can download and view log files stored in the Dolby Voice Huddle system or on Dolby Conferencing Console.

If logs are located here	Do this
Dolby Conferencing Console	Use the Dolby Conferencing Console web interface. See the .
Dolby Voice Huddle flash memory  Note: All logs originate from the device flash memory. The flash memory always has the most recent log entries, some of which may not have been uploaded yet.	1. Access the Dolby Voice Huddle web interface. 2. Select Tools, and then click Download Logs. 3. Save the <i>tar.gz</i> file, and unpack it with your preferred decompression utility. The TAR archive contains one or two log files, named <i>message</i> and <i>message.0</i>. The file named <i>message</i> contains the latest entries. 4. Open the log files with your preferred text editor.

Related information

[Event logging](#) on page 66

[Log levels](#) on page 67

[Audio logging](#) on page 67

[syslog logging](#) on page 68

[Enabling HDMI logging](#) on page 69

6.7 Initiating Reprovisioning

The most common reasons to initiate reprovisioning of Dolby Voice Huddle are to incorporate a configuration change (such as new IP PBX settings) and to update the system software.

About this task

There are several methods for initiating reprovisioning:

- With Dolby Conferencing Console
- With the web UI
- With a diagnostic shell command
- By rebooting
- By enabling scheduled provisioning

 **Note:** Check with your conferencing service provider to see if their application initiates reprovisioning by regularly rebooting the system.

Initiating reprovisioning with Dolby Conferencing Console

Initiating reprovisioning from Dolby Conferencing Console is how you push software updates to the devices. See the *Dolby Conferencing Console Administrator's Guide* for details.

Initiating reprovisioning with the web interface

1. Access the Dolby Voice Huddle web interface.
2. Go to **Tools** and click the **Provision** button.

Initiating reprovisioning with a diagnostic shell command

1. Access the diagnostic shell.
2. Enter this command:

```
system provision
```

Initiating reprovisioning by rebooting

You can initiate reprovisioning by rebooting the Dolby Voice Huddle.

Enabling scheduled provisioning

For scheduled provisioning, you define a daily time window by specifying an on-the-hour start time and a duration in minutes. Reprovisioning happens randomly within this window.

Enable scheduled provisioning using one of these methods:

- Enable with the web interface.
 1. Access the Dolby Voice Huddle web interface.
 2. Under the **Settings** tab, go to **Provisioning > Scheduled Provisioning**.
 3. Turn on **Enable Scheduled Provisioning**, and enter a start time and interval duration.
 4. Use the sliders to set a **Scheduled Reprovisioning Start Time (hour)** and a **Scheduled Reprovisioning Interval Duration (minutes)**.

For example, setting the start time to 23 and the interval duration to 90 defines a 90-minute window starting at 11:00 p.m. Reprovisioning takes place at a random time between 11:00 p.m. and 12:30 a.m.

- Enable by setting configuration parameters.
 1. Access the diagnostic shell.
 2. Set these parameters:
 - `Provisioning.Scheduled.Enable`
 - `Provisioning.Scheduled.StartTime`
 - `Provisioning.Scheduled.Duration`

See the **Help** section of the web interface for details on these parameters.

6.8 Enabling power saving mode

You can enable power saving mode and specify the number of minutes for which the system will idle before turning the TV off. By default, power saving is disabled.

About this task

There are two power saving settings: **Off Hours Idle Timeout** and **Work Hours Idle Timeout**. The working hours are defined by the start/end times and days of the work weeks. If power saving is not set within the specified work hours, then it is automatically set to **Off Hours Idle Timeout**.

Procedure

1. Access the Dolby Voice Huddle web interface.
2. Expand **Settings > User Preferences > Power Saving**.
3. Specify the **Off Hours Idle Timeout** and the **Work Hours Idle Timeout**.
4. Specify the **Work Week Start Day** and **Work Week End Day**.
5. Specify the **Work Day Start Time** and **Work Day End Time**.
6. Click **Save**.

6.9 Changing network settings

You might need to change network settings after provisioning is complete.

About this task

Using the web interface, the controller interface, or the diagnostic shell, you can change several network settings, including:

- 802.1X credentials
- NTP server address
- VLAN number
- Proxy settings
- Services

If DHCP network configuration is off, you can also set basic IP settings such as IP address, gateway, DNS server, and so on.

 **Note:** Many network setting changes cause a reboot to occur.

Procedure

- Change network settings with the web interface:
 - a) Access the Dolby Voice Huddle web interface.
 - b) Expand **Settings > Network**.

- c) If **DHCP Network Configuration Method** is off, change basic IP settings. Or, expand one of these sections and enter your changes:
 - IEEE 802.1X
 - NTP
 - Virtual LAN
 - Proxy
 - Services
- Change network settings with the controller UI:
 - a) Navigate to the Dolby Voice Huddle settings menu.
This may be different for each service provider.
 - b) Tap **Administrative Settings**.
 - c) Enter the admin password and tap **Enter**.
 - d) Tap **Network Configuration** and tap one of these menu choices:
 - IP Network
 - Ethernet
 - Network Time Server
 - 802.1X
 - Secondary VLAN
- Change network settings with the diagnostic shell and the **configure** command:
 - a) On the **Help** tab of the web interface, click the link to view the Dolby Voice Huddle configuration parameters, and then click the **Network** link to scroll to the network-specific parameters.
 - b) Locate the configuration parameter that you need to change.
 - c) Open the diagnostic shell and log in.
The default credentials are *admin* and *1739*. We recommend that you change the administrative password.
 - d) Enter the **configure** command to change the network setting.
For example, enter a command similar to this to change the 802.1X authentication password:


```
configure Network.Eapol.Password y6ttr4ub
```

6.10 Enabling Privacy block

When privacy block is disabled (the default), the Dolby Voice Huddle system can share usage data with your conferencing service provider. You can enable or disable privacy block at any time from the controller UI, web interface, or command-line interface (CLI).

About this task

Privacy block is not a provisionable feature. You must enable or disabled it one system at a time. The name of the configuration parameter that controls this feature is `Provisioning.Server.PrivacyBlock`.

From the Dolby Voice Huddle web interface:

1. From the **Settings** tab, expand the **Provisioning** section.
2. Expand the **Server** section.
3. Turn on **Disable Privacy-related Uploads**.

From the controller user interface:

1. Access the Dolby Voice Huddle setting menu.
2. Tap **Administrative Settings**.
3. Enter the administrative password.

4. Tap **Provisioning Server**.

5. Scroll down and enable **Disable Privacy-Related Uploads**.

The `Provisioning.Server.PrivacyBlock` configuration parameter is now `true`. The system cannot share data with your conferencing service provider's provisioning service.

6.11 Call and audio statistics

Through the Dolby Voice Huddle web interface, you can view call statistics for active audio calls.

To view call statistics:

- Access the web interface and select **Status > Call Statistics**.

The call statistics page shows:

Call information:

- Calling party first and last name
- Calling party address (extension)
- Call duration in minutes and seconds
- Type of call (IP PBX or conferencing service provider)
- Audio codec type

Network statistics

- Packet count
- Average jitter
- Packet loss rate
- Packet loss burstiness (rate of audio packet losses occurring in a burst)
- Concealed packet count
- Jitter histogram

Audio diagnostics

- Average automatic gain control gain
- Voice capture level
- Voice rendering level
- Signal-to-noise level
- Signal-to-echo level
- Average round-trip time (RTT)
- Signal-to-noise level too low
- Signal-to-echo level too low
- Low input power detected
- Microphone input clipping

6.12 Enabling WebRTC tracing

Enable Web Real-Time Communication (WebRTC) tracing when instructed to do so by your conferencing service provider or Dolby Customer Support.

About this task

When enabled, WebRTC traces can be uploaded to the provisioning server. With Dolby Conferencing Console, traces are uploaded automatically. Using the Dolby Conferencing Console web interface, you can download the trace files from the **Files** tab for each system.

Procedure

1. Access the Dolby Voice Huddle web interface.
2. In the **Settings** tab, expand **Logging > WebRTC**.
3. Turn on **Debug Trace**.

6.13 Rebooting Dolby Voice Huddle

You might need to reboot Dolby Voice Huddle if the system becomes unresponsive, if you have made certain network configuration changes, or if requested to do so by Dolby Customer Support or your conferencing service provider customer support.

About this task

There are several methods for rebooting:

- With the controller UI
- With the web UI
- With a diagnostic shell command (if connection over the network is possible)

Rebooting with the controller UI

1. Navigate to the Dolby Voice Huddle settings menu.

This may be different for each service provider.

2. Tap **General Settings > Reboot System**.
3. Tap the confirm button to reboot the system.

Rebooting with the web interface

1. Access the Dolby Voice Huddle web interface.
2. Go to **Tools** and click the **Reboot** button.
3. Click **Confirm** to proceed with the reboot.

Rebooting with the diagnostic shell

1. Open the Dolby Voice Huddle diagnostic shell.
2. Log in with administrator credentials. (Default credentials are *admin* and *1739*.)
3. Type this command and press Enter:

```
system reboot
```

Results

Reboot proceeds even if a call is active.

6.14 Resetting to factory defaults

Resetting Dolby Voice Huddle to factory defaults clears the user data, such as the recent calls list and contact directory, resets user preferences, removes log files and cached data, and restarts the out-of-the-box (OOB) provisioning process.

About this task

You can also reset to factory defaults to recover from a situation in which the system becomes unresponsive.

There are a few methods for resetting to factory defaults:

- With the reset button
- With the controller UI

- With the web interface
- With a diagnostic shell command (if connection over the network is possible)

Resetting with the factory reset button.

1. Power down the unit.
2. Press and hold down the reset button on the back of the unit.
3. Power up the unit and wait for 5 seconds. The red LED should light up on the front panel.
4. To reset to factory defaults, release the button. The LED light should blink to acknowledge the choice.
5. Hold down the button for 3 more seconds for recovery mode. The LED light should switch to blue.
6. To roll back to previous bank, release the button. The LED light should blink to acknowledge the choice.

Rebooting with the controller UI

1. Navigate to the Dolby Voice Huddle settings menu.

This may be different for each service provider.

2. Tap **Administrative Settings > Factory Reset**.
3. Tap the confirm button to reset the system.

Resetting with the web interface

1. Access the Dolby Voice Huddle web interface.
2. Go to **Tools** and click the **Factory Reset** button.

Resetting with the diagnostic shell

1. Open the Dolby Voice Huddle diagnostic shell.
2. Log in with administrator credentials. (Default credentials are *admin* and *1739*.)
3. Type this command, and press Enter:

```
system factoryReset
```

4. When prompted for confirmation, press Enter.

Results

The system reboots to complete the reset process even if a call is active.

6.15 Updating the Dolby Voice Huddle software

This section describes how to update the Dolby Voice Huddle software.

In a plug and play provisioning environment, your conferencing service provider manages all Dolby Voice Huddle software updates.

In an automatic provisioning environment, you obtain the latest software image from your conferencing service provider, upload it to Dolby Conferencing Console, and then install it by either pushing the software image from Dolby Conferencing Console, or by initiating reprovisioning from the Dolby Voice Huddle system.

After the initial (out-of-the-box) provisioning, reprovisioning can occur at regular intervals or on demand. If a software update is available on Dolby Conferencing Console, reprovisioning causes the update to be installed.

For large-scale installations with automatic provisioning and Dolby Conferencing Console, we recommend pushing the software update from Dolby Conferencing Console.

See the *Dolby Conferencing Console Administrator's Guide* for details on uploading software images and on pushing software updates to devices.

6.16 Configuring HDMI input audio

Dolby Voice Huddle allows you to stream HDMI input audio with a conference uplink stream for webRTC calls or SIP video calls. By default, HDMI input audio is enabled.

To stream HDMI input audio into conference uplink audio for SIP video calls, the telephony application handles the streaming of HDMI input audio when you select the **HDMI share** option during the call.

To stream HDMI input audio into conference uplink audio for webRTC calls, the webRTC application must select the audio device exposed as **Dolby HDMI Input Audio**.

The WebRTC application can select **Dolby HDMI Input Audio** in-call or out-of-call scenario. In an out-of-call scenario, HDMI input audio is rendered locally. In an in-call scenario, HDMI input audio is additionally streamed into conference uplink audio.

You can see the status of the HDMI Input Audio streams and respective pulseaudio device status through the Dolby Voice Huddle UI.

1. Log into the Dolby Voice Huddle UI.
2. Navigate to **Status > Dolby Voice Huddle > Device > Alarms and status**.

6.16.1 Disabling HDMI input audio

Dolby Voice Huddle allows you to stream HDMI input audio with a conference uplink stream for webRTC calls or SIP video calls. By default, HDMI input audio is enabled.

About this task

You can disable HDMI input audio through the Dolby Voice Huddle web UI.

Procedure

1. Log into the Dolby Voice Huddle web UI.
2. Navigate to **Settings > Features**.
3. Turn off **Stream HDMI Input Audio To Remote End**

6.17 Finding the system serial number and software version

You need the system serial number and software version when reporting a problem.

Procedure

1. Navigate to the Dolby Voice Huddle settings menu.
This may be different for each service provider.
2. Tap **About > System**.
The software version and serial number are displayed.

6.18 Finding the serial number and software version

You need the Dolby Voice Camera serial number and software version when reporting a problem.

Procedure

- 1.
2. Navigate to the Dolby Voice Huddle settings menu.
This may be different for each service provider.
3. Tap **About > Status > Camera**.

Troubleshooting

If you encounter an issue with Dolby Voice Huddle, we recommend that you attempt basic troubleshooting before you contact Support. The information in this chapter describes likely causes of issues and corrective actions you can take to resolve them. Follow the available troubleshooting procedures to review logs and error messages, run diagnostics, and possibly upgrade the system software.

- [Report to Dolby](#)
- [Service QR code](#)
- [Web administration portal access over Wi-Fi](#)
- [Enabling screenshots](#)
- [Dolby Voice Camera issues and solutions](#)
- [Dolby Voice Huddle system issues and solutions](#)
- [Display issues and solutions](#)
- [HDMI input issues](#)
- [CEC issues](#)

7.1 Report to Dolby

Report to Dolby allows you to easily send system logs to Dolby customer support.

About this task

You can send log files to *diagnostics.dolby.com*. Your service provider must work with Dolby customer support to retrieve this data.

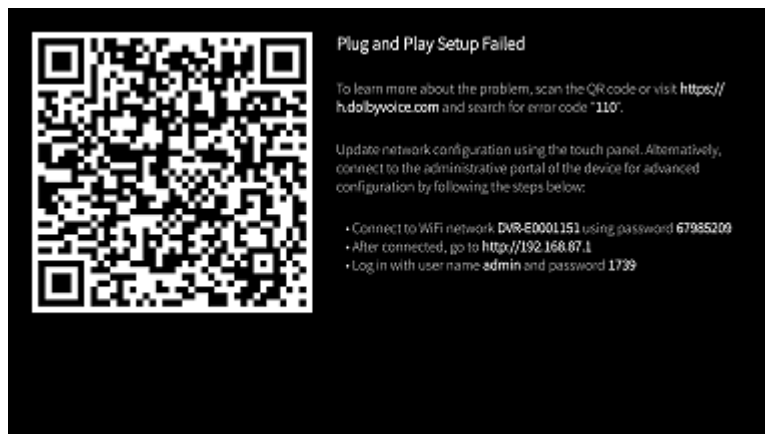
Procedure

1. Navigate to the Dolby Voice Huddle settings menu.
This may be different for each service provider.
2. Tap **General Settings > Report Problem**.
The **Report Problem** screen displays.
3. Provide details regarding the issue you experienced.
4. Tap **Yes** to send diagnostic information to Dolby.
5. Tap **Send**.

7.2 Service QR code

The service QR code provides Dolby Voice Huddle system information and status during out-of-the-box configuration.

If plug-and-play capability fails during the out-of-the-box configuration, a service QR code displays on your TV screen.



You scan the service QR code using your mobile device, and a URL to a troubleshooting page is provided. Dolby may automatically direct you to your service provider's support website, based on the serial number of the Dolby Voice Huddle.

The troubleshooting page provides the following information:

- Serial number and version
- Problem description and resolution
- Request for additional support
- Advanced system information

7.3 Web administration portal access over Wi-Fi

Dolby Voice Huddle provides access to the administrative portal over Wi-Fi during out-of-the-box configuration.

If plug-and-play capability fails during the out-of-the-box configuration, the Wi-Fi access point is automatically enabled, allowing you to connect to Dolby Voice Huddle.



You can connect to Wi-Fi through a mobile device or any Wi-Fi enabled device using the Wi-Fi network name and password displayed on the TV, along with the service QR code.

Once connected, you can launch a web browser and log into the administrative portal. The administration web portal IP address and login credentials are displayed on the TV, along with the service QR code.

You can download log files through the Dolby Voice Huddle web administration portal.

In addition, you can modify the network configuration, as well use the web administration portal.

Wi-Fi is disabled when out-of-the-box configuration is complete.

7.4 Enabling screenshots

You can enable screenshots in Dolby Voice Huddle.

Prerequisites

Contact Dolby Customer Support to obtain the power user key to enable screenshots in Dolby Voice Huddle.

Procedure

1. Apply the power user key through SSH command-line.

```
DolbyCli>system poweruser key A123:4567:BC8D:910E
Your device serial ID is 100
Power user key is set to: A123:4567:BC8D:910E
Validating ... Key is valid.
Current time is: Mon Jun 03 04:21:56 2019
Key expires: Wed Jul 03 14:14:07 2019
```

2. Enable the system poweruser devtool.

```
system poweruser devtool enable
```

3. Log into the Dolby Voice Huddle web UI.
4. Navigate to **Status > Dolby Voice Huddle > Device > Power User** and verify that **Screenshot Enabled** is set to true.
5. Navigate to **Tools > TV Screenshot** through the web-UI and you should be able to see Dolby Voice Huddle display image.

7.5 Dolby Voice Camera issues and solutions

Use this information for likely causes and corrective actions for Dolby Voice Camera issues.

Issue	Solution
Video artifact with slow moving shaded band is visible.	This occurs when HDR is enabled in rooms with dimmable LED lights. Disable HDR.
This message shows on the display: Cannot complete the Dolby Voice Camera software update	The Dolby Voice Huddle Controller user interface displays the reason why the software update failed. Check for an alarm on the Status screen on the controller user interface. You can check the software update status of the camera in the web interface by going to Status > Dolby Voice Camera > Status

7.6 Dolby Voice Huddle system issues and solutions

Use this information for likely causes and corrective actions for the Dolby Voice Huddle system.

7.6.1 Power and start-up issues

For power and start-up issues, see these possible solutions.

Issue	Solution
The Dolby Voice Huddle system appears to have no power.	Determine whether the problem is caused by the system or the AC outlet. Do one of the following: • Check the power light indicator is on. It lights up when the unit is receiving power. • Ensure that the display (TV or monitor) is powered on and connected to the HDMI port. • Ensure that the power adapter is attached securely. • Ensure that the A/C outlet is functioning. • Check for a faulty power adapter. Replace the adapter and try again.

7.6.2 Dolby Voice Huddle power saving issues and solutions

Use this information for likely causes and corrective actions for Dolby Voice Huddle power saving issues.

Issue	Solution
Disable power saving	1. Access the Dolby Voice Huddle web interface. 2. Expand Settings > User Preferences > Power Saving. 3. Disable power saving parameters. 4. Click Save.
TCL TV goes to the home screen when the system wakes	TCL TV has a feature that launches the home screen or last used TV input or specific input. By default home screen is enabled. This setting can be changed.

7.6.3 Network access issues

Here are possible solutions to network access issues.

Issue	Solution
The web interface Status tab indicates an alarm, and an alarm icon appears for the network status Status > Dolby Voice Huddle > Device > Alarms and status . Position your mouse pointer over the alarm icon to view the network status code.	If one of these status codes appears, check whether the switch is correctly set up and if the Ethernet cable is acceptable: - Link is dormant: Network is pending authentication. - Link is down. - Unknown link error: There is an unknown network error. If this status code appears, check the network status: FAIL_DHCP: Failed. If this status code appears, check whether another device on the network is using the same IPv4 address: mac-address: Another device on the network is using the same IPv4 address. If one of these status codes appears, check whether the default gateway is up: - Default gateway is unreachable. - Default gateway is unconfigured. If one of these status codes displays, check whether the DNS servers are up: - DNS servers are unreachable. - DNS servers are unconfigured. - Only secondary DNS server is reachable: The primary DNS server is unreachable, whereas the secondary DNS server is reachable. If one of these status codes displays, check that the dynamic DNS is correctly configured on the DNS servers and the DHCP server: - DNS hostname unresolvable: The host name is not resolvable. - DNS servers are unconfigured.

7.6.4 Authentication issues

Here are possible solutions to authentication issues.

Issue	Solution
You forgot the administrative password.	Perform a factory reset. The default administrative password (1739) is restored. After factory reset, you must reenter user preferences, such as language selection, date format, brightness, auto-answer, and so on.

7.6.5 HDMI input issues

For HDMI input issues, see these possible solutions.

Table 7: HDMI input troubleshooting issues

Issue	Solution
HDMI input detection issues	- Check the HDMI Input Detected parameter. It should be set to <i>TRUE</i>. - Check the detection errors using the HDMI Input Error parameter - Validate the device type and name using the HDMI Input Device Type parameter.
Audio streaming issues	Verify that the status of these parameters are green HDMI Input Audio Device Status HDMI Input Audio Stream Status If the status for both parameters is not green, check the audio output settings on the HDMI input device and confirm the audio is routed to HDMI. Further isolate the issue by connecting the HDMI input source directly to the TV. If HDMI input audio is not transmitted in the meeting but renders fine without the meeting, consult your conference service provider.
Video streaming issues	Check the video stream status using HDMI Input Stream Active, it should be <i>TRUE</i> when active HDMI Input Stream Rate is reported.
Log collection for detection issues	- Log-in to web UI. - Navigate to Logging > HDMI Input. - Set Enable HDMI Input Detection Logs to <i>On</i>. - Reproduce issue, download and share logs with support team member.
Video capture issues	- Log-in to web UI. - Navigate to Logging > HDMI Input. - Set Enable HDMI Input Detection Logs to <i>On</i>. - Set Enable HDMI Input Capture Logs to <i>On</i>. Reproduce the issue, download and share the logs with a support team member.

You can check the HDMI status through the UI, by navigating to **Status > Dolby Voice Huddle > Device > Alarms and status**.

You can check the HDMI logs through the UI, by navigating to **Settings > Logging > HDMI Input**

7.6.6 Software update issues

Here are solutions for Dolby Voice Huddle software update issues.

Issue	Solution
The message Invalid signature on software image file appears on the controller .	The image is signed by Dolby and the system performs a validation upon software upgrade. If you see this message, you are using an unauthorized image. Contact Dolby Customer Support if you received the image from Dolby. Otherwise, contact your conferencing service provider.
The message Invalid software image file appears on the controller .	After downloading the image, the system performs a checksum validation. This message appears if the validation failed. Reasons for this include transmission errors or a corrupt image file. If the image file came from Dolby, contact Dolby Customer Support. Otherwise, contact your conferencing service provider. If the issue persists, check with your organization's IT department to investigate any network transmission errors.
The message Failed to complete software update for an unknown reason appears on the controller .	During the software upgrade, a possible flash error was detected. Retry the software update. If the message appears again, contact Dolby Customer Support.

7.6.7 Video quality issues

Use this information for likely causes and corrective actions for video quality issues.

Issue	Solution
No image out of the camera in the preview (picture-in-picture (PiP)) or Camera not connected message.	Refer to the section on Dolby Voice Camera issues and solutions.
Remote participants report no video signal, but video preview (PiP) looks good.	Likely problem with the conference service provider application. Reboot the system and report the problem to the service provider.
Remote participants say video looks grainy, noisy, or out of focus.	Adjust lighting so that the room is well lit. If that does not remedy the issue, then adjust image settings as follows: 1. End the call and access the Settings screen. 2. Select Dolby Voice Camera to get a full-screen preview of the outgoing video signal. 3. Try these adjustments: • Adjust brightness, color intensity, and contrast. Avoid over-saturated skin tones. • If the room has no outside-facing windows, turn HDR off. Otherwise, if the image is washed out or faces are dark due to bright daylight coming in through a window, turn HDR on. In a well-lit room with these adjustments, the image should be sharp at a distance of one to three meters. If the preview still shows an out-of-focus image, return the camera for a replacement.
After making above adjustments, video preview looks good but remote participants say video is still poor quality.	There may not be enough up-link bandwidth, so the Dolby Voice Huddle system has automatically reduced transmission resolution to cope with the network. Consult your conferencing service provider for more information regarding minimum network requirements.

Issue	Solution
Poor video quality coming from the remote room.	Ask the remote conference participants to adjust the lighting in the room and adjust the image settings on their camera. If this does not resolve the issue, there may not be enough down-link bandwidth, so the remote Dolby Voice Huddle system has automatically reduced transmission resolution to cope with the network. Consult your conferencing service provider for more information regarding minimum network requirements.
Artifacts on the screen, such as flickering colored dots.	Replace the high-definition video cable to the display. We recommend using the cable included in the packaging. If you use your own cable, ensure that it is a high-speed cable.

7.6.8 Corporate directory issues

Here are solutions for corporate directory issues.

Issue	Solution
Cannot connect to the LDAP directory. The system status shows there is a directory error.	The possible reasons for the failure are: • An incorrect LDAP server IP address • Incorrect LDAP server access credentials • An incorrect or missing LDAP server root CA certificate (necessary when accessing the LDAP server using Secure Sockets Layer (SSL)) In the Dolby Voice Huddle web interface, go to . Find LDAP connected. Status > Dolby Voice Huddle > Device > Alarms and status. A red circle indicates an LDAP error. Position your mouse pointer over the red circle to see the specific error that occurred.

7.6.9 Fatal error issues

Here are possible solutions to fatal error issues.

Issue	Solution
This message appears on the display: The services application has stopped responding. Press Debug to stop rebooting.	You can choose to debug the problem. This stops the system from performing the automatic reboot. If you do not tap Debug, the system automatically reboots after five seconds. If the system is configured to upload log files, the system uploads the core dump (if there is any) to the provisioning server during the next reboot. The core dump file name is <i>SN-timestamp-core.appname</i>. For example, <i>D0025416-1402090840-core.services_app</i> is a core dump generated from the <i>services_app</i> application on the device numbered D0025416. Attach the uploaded log files as well as the core dump file, and create a bug report for Dolby Customer Support to perform further analysis.  Note: Uploading a core dump during booting prolongs boot time.
The controller user interface continues to display the Dolby logo and does not display the home screen, even after 90 seconds.	A combination of configuration and software issues have left the system in a state from which it cannot recover. If the reboot fails, then perform a factory reset. If the reboot still stalls at the Dolby logo, contact Dolby Customer Support.

7.7 Display issues and solutions

Use this information for likely causes and corrective actions for display (TV or monitor) issues.

Issue	Solution
No picture	• Ensure that the display is powered on. • Ensure that Dolby Voice Huddle is powered on. (The power indicator light should be lit up.) • Ensure that the display is connected to the high-definition video ports. • Disconnect and reconnect the high-definition video cable • Reboot • Try another high-speed (CAT2) high-definition video cable • Try another display

7.8 HDMI input issues

For HDMI input issues, see these possible solutions.

Table 8: HDMI input troubleshooting issues

Issue	Solution
HDMI input detection issues	• Check the HDMI Input Detected parameter. It should be set to <i>TRUE</i>. • Check the detection errors using the HDMI Input Error parameter • Validate the device type and name using the HDMI Input Device Type parameter.
Audio streaming issues	Verify that the status of these parameters are green • HDMI Input Audio Device Status • HDMI Input Audio Stream Status If the status for both parameters is not green, check the audio output settings on the HDMI input device and confirm the audio is routed to HDMI. Further isolate the issue by connecting the HDMI input source directly to the TV. If HDMI input audio is not transmitted in the meeting but renders fine without the meeting, consult your conference service provider.
Video streaming issues	Check the video stream status using HDMI Input Stream Active, it should be <i>TRUE</i> when active HDMI Input Stream Rate is reported.

Table 8: HDMI input troubleshooting issues (continued)

Issue	Solution
Log collection for detection issues	1. Log-in to web UI. 2. Navigate to Logging > HDMI Input. 3. Set Enable HDMI Input Detection Logs to <i>On</i>. 4. Reproduce issue, download and share logs with support team member.
Video capture issues	1. Log-in to web UI. 2. Navigate to Logging > HDMI Input. 3. Set Enable HDMI Input Detection Logs to <i>On</i>. 4. Set Enable HDMI Input Capture Logs to <i>On</i>. Reproduce the issue, download and share the logs with a support team member.

You can check the HDMI status through the UI, by navigating to **Status > Dolby Voice Huddle > Device > Alarms and status**.

You can check the HDMI logs through the UI, by navigating to **Settings > Logging > HDMI Input**

7.9 CEC issues

For CEC connection issues, see these possible solutions.

Issue	Solution
If the Dolby Voice Huddle is connected to the TV through a High-Definition Multimedia Interface switch, the CEC control feature may not work.	Perform a diagnostic test to verify if the setup is working.
The CEC controller on the Dolby Voice Huddle may run into a bad state and stop sending or receiving commands.	Full power cycle of the Dolby Voice Huddle restarts controller and it starts working as normal.
Some TVs may behave unexpectedly for CEC controls.	For SMART TVs, update the TV firmware to the latest version before conducting the test. This should resolve the CEC control problems.

Use these commands if dual display is being used and the CEC control is not working as expected:

Command	Description
<code>system hdmi-cec list-firmware</code>	Shows the current firmware version, as well as firmware available for updating.
<code>system hdmi-cec update-firmware</code>	Automatically updates firmware to latest version. If the firmware is up-to-date, an error will be returned.
<code>system hdmi-cec update-firmware <fw_file_to_flash></code>	Writes firmware from the specified file and disregards the version information. This can be used when downgrading or manually updating to a specific version.

Error Messages

These tables list error messages, possible causes, and suggested actions.

When an error occurs, users will see warning icon on the home screen and an informative message when the warning is first detected.

- [Dolby Voice Console error messages](#)
- [Provisioning service error messages](#)
- [Dolby Voice Huddle application error messages](#)
- [Dolby Voice Huddle touch controller application error messages](#)

8.1 Dolby Voice Console error messages

These are possible solutions to Dolby Voice Console errors when deploying Dolby Voice Huddle with the plug and play provisioning method.

These messages appear on the Dolby Voice Huddle screen.

Error code	Message displayed	Solution
101	Plug and play setup failed because the system has failed 802.1X authentication, which is required by your network. You can modify 802.1X settings in the system according to your network security policy. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the system has failed 802.1X authentication, which is required by your network. You can modify 802.1X settings in the system according to your network security policy. 802.1X, also known as port guard, configuration on the system often requires the security team in your IT organization to provide help and guidance. If you are not sure about the 802.1X policy, consult your security team for more information. You can modify basic 802.1X settings by tapping the Next button on the Dolby Voice Huddle to navigate to the network settings screen. You can toggle between 802.1X authentication methods, turn off 802.1X authentication, enter 802.1X authentication credentials, or accept server certificates to get the device authenticated against your network. You may also need to import a server certificate into the system through its web user interface.
102	Plug and play setup failed because the system was unable to obtain a valid IP address. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the system was unable to obtain a valid IP address for communicating with the Dolby Voice Console . The system needs an IP address that can reach Internet in order to function properly. If there are multiple network ports in the room, you can try to plug the main network interface to a different port. If your network uses VLAN, the system may need to be configured for a specific VLAN. The system supports manual VLAN configuration. You can modify the VLAN settings by tapping the Next button on the Dolby Voice Huddle . You may need to consult your IT help desk for assistance.
103	Plug and play setup failed because a DNS server was not configured for the system. Contact your IT help desk for assistance.	The plug and play provisioning method failed because a DNS server was not configured for the system. The system requires a valid DNS server to resolve the host name to the IP address on the Internet. If your DHCP server did not supply a DNS configuration to the system, you can configure the DNS server manually on the . Follow the instructions on the Dolby Voice Huddle to provide the DNS server settings; for example you can enter the public DNS server 8.8.8.8.

Error code	Message displayed	Solution
104	Plug and play setup failed because the system cannot complete proxy auto detection. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the system was unable to download the PAC file to complete the setup. The system can automatically detect proxy server on your network. It has detected a PAC file, but was unable to download the file to complete the auto discovery process. Follow the instructions on the Dolby Voice Huddle to disable WPAD. You can either leave the PAC URL field blank, or enter a valid URL for the system to download the PAC file.
105	Plug and play setup failed because the proxy on your network requires authentication. Provide the proxy authentication credentials on the next screen.	The plug and play provisioning method failed because the proxy on your network requires authentication. If your proxy server requires authentication, the system must be configured with authentication credentials. If you do not know what credentials to use, contact your IT organization's security team. If you already know the authentication credentials, tap the Next button and move to the next screen to supply proxy authentication credentials and try again.
106	Plug and play setup failed because of an unknown proxy error. Contact your service provider for assistance.	The plug and play provisioning method failed because of an unknown proxy error. Contact your conferencing service provider for help.
107	Plug and play setup failed because the Dolby Voice Console (https://console.dolbyvoice.com) host name is not resolvable. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the Dolby Voice Console host name is not resolvable. The system needs to connect to the Dolby Voice Console to discover the provisioning service. This is required for the plug and play provisioning method. The DNS server configured on system must be able to resolve the Dolby Voice Console host name. Tap the Next button to review the network configuration. You can manually override the DNS server settings on the next screen. Configuring the public DNS server 8.8.8.8 may help resolve this issue.
108	Plug and play setup failed because the Dolby Voice Console (https://console.dolbyvoice.com) is unreachable from this network. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the Dolby Voice Console is unreachable from this network. The system needs to connect to the Dolby Voice Console to discover the provisioning service. This is a required step for the plug and play provisioning method setup. Your network must allow the system to connect to the Dolby Voice Console. Make sure that your firewall whitelists <i>*.dolbyvoice.com</i> to allow the plug and play provisioning method and service data traffic from the system to pass through. Consult your conferencing service provider's network preparation guidelines for all addresses required by the service.
109	Plug and play setup failed because of an unknown SSL error. Contact your service provider for assistance.	The plug and play provisioning method failed because of an unknown SSL error. The system detected an unknown SSL error and was unable to complete the plug and play provisioning method setup. Contact your conferencing service provider for help.

Error code	Message displayed	Solution
110	Plug and play setup failed because of a communication error with the Dolby Voice Console (https://console.dolbyvoice.com). Contact your service provider for assistance.	The plug and play provisioning method failed because of a communication error with the Dolby Voice Console. The system needs to connect to the Dolby Voice Console to discover the provisioning service. This is a required step for the plug and play provisioning method setup. The system experienced a communication error with the Dolby Voice Console and was unable to complete the plug and play provisioning method setup. Contact your conferencing service provider for help.
111	Plug and play setup failed because of an error in discovering provisioning information for the system. Contact your service provider for assistance.	The plug and play provisioning method failed because of an error in discovering provisioning information for the system. The system needs to connect to the Dolby Voice Console to discover the provisioning service. This is a required step for the plug and play provisioning method setup. The system experienced an error while trying to retrieve the provisioning service information from the Dolby Voice Console and was unable to complete the plug and play provisioning method setup. Contact your conferencing service provider for help.
112	Plug and play setup failed because the system's provisioning information could not be found on the Dolby Voice Console (https://console.dolbyvoice.com). Contact your service provider for assistance.	The plug and play provisioning method failed because of an error in discovering provisioning information for the system. The system needs to connect to the Dolby Voice Console to discover the provisioning service. This is a required step for the plug and play provisioning method setup. The provisioning information for the system was not found on the Dolby Voice Console and the system was unable to complete the plug and play provisioning method setup. Contact your conferencing service provider for help.
113	Plug and play setup failed because the system cannot reach the proxy server. Contact your service provider for assistance.	The plug and play provisioning method failed because the system cannot reach the proxy server. The system has detected a proxy and is trying to connect to the Dolby Voice Console using that proxy. However, the proxy server is unreachable. Follow the instructions on the Dolby Voice Huddle to disable and clear the setting for PAC configuration. If your network allows Internet access bypassing the proxy server, this may resolve the issue.
115	Plug and play setup failed because the system encountered an SSL certificate issue connecting to Dolby Voice Console (https://console.dolbyvoice.com). Contact your IT help desk for assistance.	The plug and play provisioning method failed because the system encountered an SSL certificate issue connecting to Dolby Voice Console. The system is unable to verify the identity of the Dolby Voice Console. This is usually caused by TLS/SSL intercepting proxy that is deployed in your network. You need to consult the security team of your IT organization for help. You will need to obtain the CA certificate of your intercepting proxy, and import the certificate into the system through its web user interface. See Accessing the Dolby Voice Huddle web interface on page 63 and Adding certificate authority certificates on page 35 for information on connecting to the Dolby Voice Huddle web interface using a web browser and importing the CA certificate into the device from the web interface.

Error code	Message displayed	Solution
116	Plug and play setup failed because the Dolby Voice Console is blocked by a firewall. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the Dolby Voice Console is blocked by a firewall. The system needs to connect to the Dolby Voice Console to discover the provisioning service. This is a required step for the plug and play provisioning method setup. Your network must allow the system to connect to the Dolby Voice Console. Make sure that your firewall whitelists *.dolbyvoice.com to allow the plug and play provisioning method and service data traffic from the system to pass through. For additional help, contact your conferencing service provider.
150	Plug and play setup failed because the main network interface of the is down. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the main network interface of the Dolby Voice Huddle is down. Contact your IT help desk for assistance.
151	Plug and play setup failed because the system detected another device on your network with a duplicate IP address. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the system detected another device on your network with a duplicate IP address. Contact your IT help desk for assistance.
152	Plug and play setup failed because the default gateway is unreachable. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the default gateway is unreachable. Contact your IT help desk for assistance. Verify that the DHCP assigned default gateway address on the Dolby Voice Huddle is valid. This error may be due to incorrect DHCP configuration. If your network uses VLAN for video conferencing equipment, you can enter the VLAN ID on the Dolby Voice Huddle so that the can use a different IP address and default gateway.

8.2 Provisioning service error messages

These are possible solutions to provisioning service errors when deploying Dolby Voice Huddle with the plug and play provisioning method.

These messages appear on the Dolby Voice Huddle screen.

Error code	Message displayed	Solution
201	Plug and play setup failed because the system could not connect to the provisioning server. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the system could not connect to the provisioning server. The system needs to connect to the Dolby Voice Console to discover the provisioning service. This is a required step for the plug and play provisioning method setup. Your network must allow the system to connect to the Dolby Voice Console. Log in to the system user interface to determine which provisioning server the system is configured with. Make sure that your firewall whitelists *.dolbyvoice.com to allow the plug and play provisioning method and service data traffic from the system to pass through. Consult your conferencing service provider's network preparation guidelines for all addresses required by the service.
202	Plug and play setup failed because the system encountered an SSL error while connecting to the provisioning server. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the system encountered an SSL error while connecting to the provisioning server. The system is unable to verify the identity of the provisioning server. This is usually caused by TLS/SSL intercepting proxy that is deployed in your network. You need to consult the security team of your IT organization for help. You will need to obtain the CA certificate of your intercepting proxy, and import the certificate into the system through its web user interface.
203	Plug and play setup failed because the system failed authentication with the provisioning server. Contact your service provider for assistance.	The plug and play provisioning method failed because the system failed authentication with the provisioning server. The system is attempting to connect to a provisioning server that requires authentication. If you do not know the credentials, contact your conferencing service provider.
204	Plug and play setup failed because the system cannot find the master configuration file from the provisioning server. Contact your service provider for assistance.	The plug and play provisioning method failed because the system cannot find the master configuration file from the provisioning server. The master configuration file is the first file the system downloads during the provisioning process. The system was unable to download the master configuration file. Contact your conferencing service provider for help.
205	Plug and play setup failed because the system cannot find the configuration file from the provisioning server. Contact your service provider for help.	The plug and play provisioning method failed because the system cannot find the configuration file from the provisioning server. Contact your conferencing service provider for help.
206	Plug and play setup failed because the system found an invalid master configuration file from the provisioning server. Contact your service provider for assistance.	The plug and play provisioning method failed because the system found an invalid master configuration file from the provisioning server. The master configuration file is the first file the system downloads during the provisioning process. The system found an invalid master configuration file. Contact your conferencing service provider for help.

Error code	Message displayed	Solution
207	Plug and play setup failed because the system cannot download the upgrade package metadata file. Contact your service provider for assistance.	The plug and play provisioning method failed because the system cannot download the upgrade package metadata file. The upgrade package metadata file contains software upgrade information for the system. The system uses this file to determine whether it needs to upgrade. Contact your conferencing service provider for help.
208	Plug and play setup failed because the system found an invalid upgrade package metadata file. Contact your service provider for assistance.	The plug and play provisioning method failed because the system found an invalid metadata file. The upgrade package metadata file contains software upgrade information for the system. The system uses this file to determine whether it needs to upgrade. Contact your conferencing service provider for help.
209	Plug and play setup failed because the system detected an invalid configuration from the provisioning server. Contact your service provider for assistance.	The plug and play provisioning method failed because the system detected an invalid configuration from the provisioning server. Contact your conferencing service provider for help.
210	Plug and play setup failed because the system detected an invalid or empty certificate URL from the provisioning server. Contact your service provider for assistance.	The plug and play provisioning method failed because the system detected an invalid or empty certificate URL from the provisioning server. Contact your conferencing service provider for help.

8.3 Dolby Voice Huddle application error messages

These are possible solutions to application errors when deploying Dolby Voice Huddle with the plug and play provisioning method.

Error code	Message displayed	Solution
401	Plug and play setup failed because the system failed to download the setup app due to missing files. Contact your service provider for assistance.	The plug and play provisioning method failed because the system was unable to download the setup app due to missing files. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. Contact your conferencing service provider for help.
402	Plug and play setup failed because the system failed to download the setup app due to a network error. Contact your service provider for assistance.	The plug and play provisioning method failed because the system was unable to download the setup app due to a network error. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. The system was unable to download the Dolby Voice Huddle setup app. This could mean that your firewall blocked access to your conferencing service provider's service. To determine which servers the system needs to access, see the configuration guide from your conferencing service provider. Contact your conferencing service provider for help.

Error code	Message displayed	Solution
403	Plug and play setup failed because the server rejected the request when the system tried to download the setup app. Contact your service provider for assistance.	The plug and play provisioning method failed because the server rejected the request when the system attempted to download the setup app. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. The system was unable to download the Dolby Voice Huddle setup app. Contact your conferencing service provider for help.
404	Plug and play setup failed because the system failed to download the setup app due to a network error. Contact your service provider for assistance.	The plug and play provisioning method failed because the system was unable to download the setup app due to a network error. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. The system was unable to download the Dolby Voice Huddle setup app. This could mean that your firewall blocked access to your conferencing service provider's service. To determine which servers the system needs to access, see the configuration guide from your conferencing service provider. Contact your conferencing service provider for help.
405	Plug and play setup failed because the system failed to download the setup app due to a DNS error. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the system was unable to download the setup app due to a DNS error. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. The system was unable to download the Dolby Voice Huddle setup app due to a DNS error. This error can occur if the DNS server assigned to the system is unable to resolve the server name. Contact your conferencing service provider for help.
406	Plug and play setup failed because the system failed to download the setup app due to a network error. Contact your service provider for assistance.	The plug and play provisioning method failed because the system was unable to download the setup app due to a network error. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. The system was unable to download the Dolby Voice Huddle setup app. This could mean that your firewall blocked access to your conferencing service provider's service. To determine which servers the system needs to access, see the configuration guide from your conferencing service provider. Contact your conferencing service provider for help.
407	Plug and play setup failed because the system failed to download the setup app due to a proxy error. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the system was unable to download the setup app due to a proxy error. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. The system was unable to download the Dolby Voice Huddle setup app. This error can occur if the proxy that the system has automatically discovered is unable to process the setup app download request. Contact your conferencing service provider for help.

Error code	Message displayed	Solution
408	Plug and play setup failed because the system failed to download the setup app due to an SSL error. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the system was unable to download the Dolby Voice Huddle setup app due to an SSL error. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. This is usually caused by TLS/SSL intercepting proxy that is deployed in your network. You need to consult the security team of your IT organization for help. You will need to obtain the CA certificate of your intercepting proxy, and import the certificate into the system through its web user interface.
409	Plug and play setup failed because the server rejected the request when the system tried to download the setup app. Contact your service provider for assistance.	The plug and play provisioning method failed because the server rejected the system from downloading the setup app. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. The system was unable to download the Dolby Voice Huddle setup app. Contact your conferencing service provider for help.
410	Plug and play setup failed because the system failed to load the setup app. Contact your service provider for assistance.	The plug and play provisioning method failed because the system was unable to load the setup app. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. The system was unable to download the Dolby Voice Huddle setup app. Contact your conferencing service provider for help.

8.4 Dolby Voice Huddle touch controller application error messages

These are possible solutions to touch controller application errors when deploying Dolby Voice Huddle with the plug and play provisioning method.

Error code	Message displayed	Solution
501	Plug and play setup failed because the system failed to download the setup app due to missing files. Contact your service provider for assistance.	The plug and play provisioning method failed because the system was unable to download the setup app due to missing files. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. Contact your conferencing service provider for help. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 69.
502	Plug and play setup failed because the system failed to download the setup app due to a network error. Contact your service provider for assistance.	The plug and play provisioning method failed because the system was unable to download the setup app due to a network error. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. The system was unable to download the touch controller setup app. This could mean that your firewall blocked access to your conferencing service provider's service. To determine which servers the system needs to access, see the configuration guide from your conferencing service provider. Contact your conferencing service provider for help. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 69.
503	Plug and play setup failed because the server rejected the request when the system tried to download the setup app. Contact your service provider for assistance.	The plug and play provisioning method failed because the server rejected the request when the system attempted to download the setup app. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. The system was unable to download the touch controller setup app. Contact your conferencing service provider for help. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 69.
504	Plug and play setup failed because the system failed to download the setup app due to a network error. Contact your service provider for assistance.	The plug and play provisioning method failed because the system was unable to download the setup app due to a network error. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. The system was unable to download the touch controller setup app. This could mean that your firewall blocked access to your conferencing service provider's service. To determine which servers the system needs to access, see the configuration guide from your conferencing service provider. Contact your conferencing service provider for help. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 69.

Error code	Message displayed	Solution
505	Plug and play setup failed because the system failed to download the setup app due to a DNS error. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the system was unable to download the setup app due to a DNS error. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. The system was unable to download the touch controller setup app due to a DNS error. This error can occur if the DNS server assigned to the system is unable to resolve the server name. Contact your conferencing service provider for help. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 69.
506	Plug and play setup failed because the system failed to download the setup app due to a network error. Contact your service provider for assistance.	The plug and play provisioning method failed because the system was unable to download the setup app due to a network error. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. The system was unable to download the touch controller setup app. This could mean that your firewall blocked access to your conferencing service provider's service. To determine which servers the system needs to access, see the configuration guide from your conferencing service provider. Contact your conferencing service provider for help. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 69.
507	Plug and play setup failed because the system failed to download the setup app due to a proxy error. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the system was unable to download the setup app due to a proxy error. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. The system was unable to download the touch controller setup app. This error can occur if the proxy that the system has automatically discovered is unable to process the setup app download request. Contact your conferencing service provider for help. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 69.

Error code	Message displayed	Solution
508	Plug and play setup failed because the system failed to download the setup app due to an SSL error. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the system was unable to download the touch controller setup app due to an SSL error. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. This is usually caused by TLS/SSL intercepting proxy that is deployed in your network. You need to consult the security team of your IT organization for help. You will need to obtain the CA certificate of your intercepting proxy, and import the certificate into the system through its web user interface. For additional help, contact your conferencing service provider. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 69.
509	Plug and play setup failed because the server rejected the request when the system tried to download the setup app. Contact your service provider for assistance.	The plug and play provisioning method failed because the server rejected the system from downloading the setup app. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. The system was unable to download the touch controller setup app. Contact your conferencing service provider for help. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 69.
510	Plug and play setup failed because the system failed to load the setup app. Contact your service provider for assistance.	The plug and play provisioning method failed because the system was unable to load the setup app. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. The system was unable to download the touch controller setup app. Contact your conferencing service provider for help. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 69.

Glossary

1080p

See [full high definition](#).

API

Application programming interface. A set of functions that can be used to access the functions of an operating system or other type of software.

API

Application programming interface. A set of functions that can be used to access the functions of an operating system or other type of software.

Bluetooth LE

Bluetooth low energy. A chip technology that produces radio waves to connect a device to other Bluetooth devices. The low-energy version uses less power than standard Bluetooth, but maintains a similar communication range.

CLI

Command-line interface.

CSV

Comma-separated values.

DHCP

Dynamic Host Configuration Protocol.

DNS

Domain Name System. An Internet service that translates Internet domain and host names to IP addresses and conversely. DNS automatically converts between the name entered in a web browser and the IP addresses of the web server hosting the site whose URL is entered in the web browser.

DNS SRV

Domain Name System service record. A service record (SRV record) defines the location (the host name and port number) of a server in a DNS.

DoS

Denial of service. Events that render systems on a computer network temporarily unusable. DoS can happen accidentally as the result of actions taken by network users, but is often malicious.

DSCP

Differentiated Services Code Point. An IP header field that allows a router to provide different levels of service to different types of network traffic.

DSP

Digital signal processor. A specialized microprocessor optimized for digital signal processing.

DTMF

Dual-tone multifrequency. A system that is used by touch-tone telephones to assign a specific frequency (consisting of two separate tones) to each key so that it can easily be identified by a microprocessor.

FTP

File Transfer Protocol. A network-based protocol designed for transferring data using a client-server architecture.

H.264

Also known as Advanced Video Coding (AVC), ISO/IEC MPEG-4 AVC, and ISO/IEC 14496-10:2012. An MPEG standard for video compression most commonly used for high-definition video, such as Blu-ray Disc. The standard was developed jointly by the International Telecommunication Union (ITU) and ISO/IEC MPEG.

HDMI

High-Definition Multimedia Interface. A high-speed, high-capacity format for transferring digital information and the specific hardware interface for the format.

HDR

High dynamic range.

HTTP

Hypertext Transfer Protocol. An application protocol for hypermedia information systems, and the foundation for data communication for the World Wide Web.

HTTPS

Hypertext Transfer Protocol Secure. An application protocol for secure communication over a network and the Internet that provides authentication of websites and keeps user information private.

IEEE

Institute of Electrical and Electronics Engineers.

iLBC

Internet low bit-rate codec. An open-source royalty-free narrowband speech audio coding format codec and reference implementation.

IP

Internet Protocol.

IP address

Internet Protocol address. A numerical identifier assigned to a device that is a member of a network that uses the IP for communication.

JSON

JavaScript Object Notation. A lightweight data-interchange format that is easy for humans to read and write, and for machines to parse and generate.

kbps

Kilobits per second.

LAN

Local area network.

LDAP

Lightweight Directory Access Protocol. An application protocol for querying or modifying items in corporate directories that allows sharing of information about users, devices, and applications on a network.

LLDP

Link Layer Discovery Protocol . A vendor-neutral link layer protocol for Ethernet network devices such as switches, routers, and wireless local area network (LAN) access points to announce information about themselves to other nodes on the network and store the information they discover, as defined in IEEE 802.1AB.

MAC

Multiply-accumulate. In digital signal processing, the multiply-accumulate operation is a common step that computes the product of two numbers and adds that product to an accumulator.

MAC address

Media access control address. A unique identifier assigned to a network interface for communications on a network. MAC addresses are typically assigned by the network interface manufacturer.

MPEG

Moving Picture Experts Group. An ISO/IEC working group that develops video and audio coding standards.

NAPTR record

Naming Authority Pointer record. A type of resource record in the Domain Name System (DNS) of the Internet used to map servers and user addresses in the Session Initiation Protocol (SIP). Used most frequently for Internet telephony (VoIP) services.

NAT

Network Address Translation. An Internet standard that enables a local area network (LAN) to use one set of IP addresses for internal traffic and a second set of addresses for external traffic.

NTLMv2

A Microsoft security protocol that provides authentication, integrity, and confidentiality to users. NTLMv2 is part of Windows NT LAN Manager, which is a suite of security protocols.

NTP

Network Time Protocol. A network protocol for clock synchronization on computers.

PAC

Proxy automatic configuration. A file that defines how web browsers and other user agents can automatically choose the appropriate proxy server (access method) for fetching a given URL.

PBX

Private branch exchange. A phone system that is delivered as a hosted service.

PEM

Privacy-Enhanced Electronic Mail. A file format for security certificates in email communication.

PiP

Picture-in-picture.

PKI

Public key infrastructure. A set of roles, policies, and procedures needed to create, manage, distribute, use, store, and revoke digital certificates and manage public-key encryption.

PoE

Power over Ethernet. A solution in which an electrical current is run to networking hardware over Ethernet category 5 or higher data cabling.

PSTN

Public switched telephone network. A circuit-switched telephone network operated by national, regional, or local operators.

RTCP

RTP Control Protocol. A protocol for communicating quality-of-service information about Real-time Transport Protocol (RTP) sessions.

RTP

Real-time Transport Protocol. An IP network audio and video protocol.

SBC

Session border controller. A piece of network equipment or a collection of functions that controls real-time session traffic at the signaling, call control, and packet layers as they cross a notional packet-to-packet network border between networks, or between network segments.

SIP

Session Initiation Protocol. An application-layer communications protocol used for signaling and controlling communications sessions.

SNMP

Simple Network Management Protocol. A protocol for managing IP network devices

SRTP

Secure Real-time Transport Protocol. A Real-time Transport Protocol (RTP) profile that provides encryption, message authentication, and integrity, and replay protection to the RTP data in both unicast and multicast applications.

SSH

Secure Shell protocol. An encrypted network protocol for secure data communication, remote command-line login, remote command execution, and other secure network services between two networked computers.

SSL

Secure Sockets Layer . A security protocol that works at a socket level.

TCP

Transmission Control Protocol. A communications protocol that specifies how data should be formatted, addressed, transmitted, routed, and received at the destination. Part of the Internet protocols communications suite.

TCP/IP

Transmission Control Protocol/Internet Protocol. Communications protocols that specify how data should be formatted, addressed, transmitted, routed, and received at the destination. Part of the Internet protocols communications suite.

TLS

Transport Layer Security. A cryptographic protocol designed to provide communications security over a computer network.

TLV

Type length value. Within data communication protocols, a format for encoding optional information.

UDP

User Datagram Protocol. A communications protocol that uses no handshaking dialogues to establish a connection with the remote host. UDP is a member of the IP suite.

UHD

Ultra-high definition. Ultra-high-definition television or video, with a display resolution of 3840×2160 pixels in the 16:9 aspect ratio. Also referred to as 2160p.

UI

User interface.

URI

Uniform Resource Identifier. A group of characters identifying a resource on a network (typically, the Internet).

URL

Uniform resource locator.

USB

Universal Serial Bus. A standard that defines the cables, connectors, and communications protocols used in connections between computers and electronic devices.

VLAN

Virtual LAN. Any broadcast domain that is partitioned and isolated in a computer network at the data link layer (OSI layer 2).

VoIP

Voice over Internet Protocol. A methodology and group of technologies for the delivery of voice communications and multimedia sessions over Internet Protocol (IP) networks.

WebRTC

Web Real-Time Communication. API definition drafted by the World Wide Web Consortium (W3C) that supports browser-to-browser applications for voice calling, video conferencing, and P2P file sharing without the need of either internal or external plugins.

WPAD

Web Proxy Automatic Discovery. An Internet protocol that enables a web browser to automatically connect to a cache server (or proxy server) location in a network to retrieve stored web pages more quickly by than leaving the network to request the web page from the site of the originating web server.

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